Master Equation River

A Provenance Audit Across the Human–AI Readout Programme

Version 1.6 — v1.5 unchanged, plus registrar proposals for the 24 uncoded equations and IDM/Core-Epistemic-Structure additions

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Document status. This document is an internal-literature review and provenance audit of the Human–AI Readout Programme: it arranges the equations and proposals of several works of the programme into a single river. It does **not** claim that the whole river is a universal psychological law or a single physics equation. It functions instead as a typed, multi-level dependency architecture: some segments are Readout/Domain architecture, some are definitions, some are DR theses, some are OPEN empirical hypotheses, and some are measurement protocols. The purpose is to make visible which work each segment comes from, why the segments connect, and where notation or layering still needs to be fixed before the river is frozen into canonical form.

How v1.1 differs from v1.0. This edition was checked under the glosa methodology (Section 9): every internal source is now cited to the record actually deposited on Zenodo, with its DOI (v1.0 had cited Fusion/Tunnel as "v7" when the deposited record is v8.1, and had cited RG-MEMK and Agency Potential as though separately deposited, which they are not); the retention-gate equations (35)–(36) are now given the status of a Master-edition notation proposal rather than a restatement of v8.1's own text, which does not contain this formula; the potential equation (25) uses the witnessed set of *Potential as a Readout* v2 rather than the feasible set; a disclosure has been added that H_{return} was renamed from CTSA's R_H; and a symbol glossary (Appendix A) has been added. Knowledge status K0.

Edition note (v1.2). This is an English-only edition prepared for the textbook *Written by AI. Still True.* The content is unchanged from v1.1: every section, equation (same numbering, (1)–(45)), provenance-table row, notation-collision row, fix box, reference entry, and appendix entry appears here in the same order. Passages originally written in Thai have been rendered into English with the same meaning; where the Thai and the adjacent English already said the same thing, one English sentence is kept and the merge is recorded here. Nothing has been added, nothing has been dropped, and no claim has been strengthened. Version 1.1 (Thai/English, DOI 10.5281/zenodo.22414412) remains in the record's version history as the source of record for this edition.

Edition note (v1.3). This edition adds three new segments to the unchanged v1.2 English edition: (i) Section 3.13 Topic Entry and Agenda Ownership Before the Prompt, from *How Humans Should Converse with AI* (Dialogue Conversion Protocol, DCP) [Lahtee, 2026o]; (ii) Section 3.14 Human Return → Action → World Feedback, from the same source; and (iii) Section 3.15 The World-System Layer: Human Systemic Position and Human Conversion, from *After Labour* [Lahtee, 2026m] and *The Human Conversion Imperative* [Lahtee, 2026n]. Every section, equation (1)–(45), table row, fix box, and reference entry of v1.2 is unchanged and appears here in the same order; the three new segments use continued equation numbers (46) onward and are added to the genealogy map (Table 1), the evolution-of-questions list (Section 6, items 11–13), the epistemic-status table (Table 2), and the symbol glossary (Table 5). Each new statement about a source paper is traceable to that paper's own equation or section number, cited in place. No claim is raised beyond the tier the source paper itself gives the object: most of the new objects are [DEFINITION] or accounting identities inside a stylized model, and several are explicitly [OPEN] propositions in their source. A relayed external reader's assessment of DCP world-literature convergence (2026-09-06, unverified, none of its citations independently checked here) is recorded only as a readout in Section 6; it is not treated as validation. Version 1.2 (English, same content as the deposited v1.1, DOI 10.5281/zenodo.22414412) remains in the record's version history. Knowledge status K0.

Edition note (v1.4). This edition adds one new segment to the unchanged v1.3 English edition: Section 3.16, restating RG-HCA's proactive extension [Lahtee, 2026p]. Every section, equation (1)–(66), table row, fix box, and reference entry of v1.3 is unchanged and in order; the new segment uses (67)–(78), plus (79) extending Section 5 by one sentence, and is added to Table 1, Section 6 (item 14), Table 2, and Table 5. Each new statement is traceable to RG-HCA's own equation or section, at no higher tier than RG-HCA gives: several objects are [DEFINITION], two are [OPEN], and the measurement objects are [DEFINITION, MEASUREMENT]. RG-HCA's Human Return reuse (eq. 24–25) is discussed in prose only, since it cites the same CTSA v9.1 source as H_{return} (41)–(42) under different field names, not identically. Every equation (1)–(79) is formalised in the companion Coq record (Appendix B; DOI 10.5281/zenodo.22518450). K0.

Edition note (v1.5). Every section, equation (1)–(79), table, fix box, and reference of v1.4 is unchanged and in order. This edition adds Section 10 (this river read against the programme's root spine, now coded into **Toledo** [Lahtee, 2026q]) and Appendix C (a row-by-row Toledo cross-reference for all 79 equations), both read off `registry/CANONICAL.json` and `registry/COLLAPSE.md` as they stand on 2026-09-07. Appendix B gains one pointer: DOI 10.5281/zenodo.22518450 is obsoleted by Toledo's DOI 10.5281/zenodo.22548770; its ledger table is unchanged. No claim, tier, or number already in v1.4 changes. K0.

Edition note (v1.6). v1.5 unchanged throughout ((1)–(79), all tables/fix boxes/references); additions only: (i) Core Epistemic Structure block (below); (ii) one Section 10.3 paragraph naming the Toledo/IDM ladder codes ($D, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \delta_R$; DOI 10.5281/zenodo.22644131); (iii) Appendix C's 24 uncoded rows renamed to a registrar proposal (`registry/proposals/master_river_v1_6.json`, alias `MR:eq.N`) each, no code/tier asserted; (iv) Toledo cite updated v1.0.0→v1.5.0, DOI 10.5281/zenodo.22642109. K0.

One-sentence summary. The whole internal literature can be arranged into a continuous cycle: World → Readout → Meaning → Experience → Retention → Accessibility → Live Possibility → Choice → Agency → Feedback → Changed Reader → Human–AI → Human Return → Warrant. But once a retained change has occurred, the cycle does not return to the same person: whenever the change persists, $H_{t+1} \neq H_t$, so it should be understood as a recursive river or spiral rather than a closed circle.

Core Epistemic Structure (v1.6). Core Respondent: Yaoharee Lahtee — author of every cross-referenced source; chose occurrence vs. new-entry per uncoded equation. **Interactional Expert:** None. **AI Model(s):** v1.5's disclosure names no vendor; this pass: Claude Fable 5.1 (via Claude Code) — registration, typesetting, deposit. *Non-collapse: Expert ≠ Interactional Expert ≠ AI Model (`weld/H.33.v1`, `H.30.v1`); attribution, not certification.*

1 What This Audit Found

The Master Equation River did not arise from inventing a new block of equations in one sitting. It arose from noticing that several manuscripts of the programme were each describing a different stage of the same process. Readout Genesis (and the MEMK domain contained within it) supplies the root-to-domain order and the distinction between world, readout, and meaning; Human Learning supplies representation–meaning–retrieval–constraint–competence; Human LoRA supplies selective persistence; From Problem to Hypothesis supplies history-shaped accessibility; Fusion/Tunnel supplies recursive Human–AI coupling, resistance, retention, and direction; Meaning Before Naming and Experience Is Meaning-Giving supply meaning-giving, resonance, rhythm, and transition readiness; Potential as a Readout supplies the witnessed corrigible action envelope (inheriting from the now-superseded Agency Potential note); Choice Begins Before Choice supplies the live possibility field; CTSA supplies the session-boundary Human Return audit.

What Master adds, therefore, is not any single equation on its own but the dependency closure across the papers: once every layer is laid out in order, we see that experience can change the reader, the changed reader can see the world of possibility differently, and the changed world of possibility then changes choice, action, feedback, and the ongoing Human–AI trajectory.

2 Genealogy Map: Where Each Segment Comes From

Every internal source below is cited to the record deposited on Zenodo (DOI numbers in the reference list). Rows where v1.0 had cited a document that was not separately deposited are marked † and point to the record that actually holds that content.

Table 1: Genealogy map: which deposited record supplies each segment of the river, and what it actually contributes.

Segment in Master	Primary internal source (deposited record)	What that source actually supplies
World $\rightarrow$ finite readout	Readout Genesis Standalone Synthesis [Lahtee, 2026d] (MEMK domain is in this volume †)	Root has no built-in semantic labels; retained difference precedes naming; domain translation and readout precede meaning; the reader makes the distinction accessible but does not create root truth.
Readout $\rightarrow$ meaning	Experience Is Meaning-Giving v2 [Lahtee, 2026h]; Readout Genesis [Lahtee, 2026d]	Meaning is formalized as a reader-conditioned significance relation under human state, context, and criterion/question.
Meaning $\rightarrow$ experience	Readout Genesis [Lahtee, 2026d] eq. (14); Experience v2 [Lahtee, 2026h]	Genesis gives $E_{i,n} = \Phi_E(x_{i,n}, \mu_{i,n}, \kappa_{i,n}, c_n)$; Experience v2 makes the explicit thesis that experience is phenomenon-as-meaningfully-read.
Experience $\leftrightarrow$ naming	Meaning Before Naming [Lahtee, 2026g]; Experience v2 [Lahtee, 2026h]	Allows E_n, μ_n to arise before stable naming, and adds the recursive correction that naming can restructure later meaning/experience.
Experience $\rightarrow$ retained future reader	Experience Is the Human LoRA [Lahtee, 2026c]; Readout Genesis [Lahtee, 2026d]	Experience is not equal to adaptation; only a selected residue is retained/consolidated and goes on to change the future interpretive/action operator.
External–internal congruence	Experience v2 [Lahtee, 2026h]	Resonance is formally defined as the congruence between the current externally prompted experience and the retained internal experiential organization (the earlier definition of resonance in [Lahtee, 2026g] is superseded).
Ordered recurrence / rhythm	Experience v2 [Lahtee, 2026h]	Rhythm is structured recurrence, spacing, interruption, emphasis, boundary, and return over ordered history; it is not meaning and not physical resonance.
History $\rightarrow$ re-entry	From Problem to Hypothesis [Lahtee, 2026e]	Semantic momentum is a finite-history proxy: a route just traversed may be easier to re-enter; this is an Open hypothesis.
Attraction + momentum + switching cost $\rightarrow$ accessibility	From Problem to Hypothesis [Lahtee, 2026e]	Supplies the history-shaped accessibility kernel and stresses that accessibility/attraction/momentum are not warrant.
Accumulation $\rightarrow$ transition readiness	Readout Genesis [Lahtee, 2026d] (MEMK †); Experience v2 [Lahtee, 2026h]	Semantic potential, informational work, and barrier crossing are used as a domain-level transition topology; must not be read as physical energy.
Feasible $\rightarrow$ live possibility	Choice Begins Before Choice [Lahtee, 2026j]	Places a live possibility field between feasible capacity and overt choice.
Live profile / live set	Choice Begins Before Choice [Lahtee, 2026j]	Converts the live option from a binary set into a distribution of practical accessibility and a measurement-specific threshold.
Live $\rightarrow$ choice $\rightarrow$ action $\rightarrow$ observation	Choice Begins Before Choice [Lahtee, 2026j]; Potential as a Readout [Lahtee, 2026i]	Separates possible, feasible, live, chosen, enacted, and observed, and separates evaluator visibility from underlying capacity (NC-78: potential $\neq$ exercised $\neq$ observed).
Effective corrigible agency	Potential as a Readout v2 [Lahtee, 2026i] (replacing the Agency Potential note †)	Agency is a task-relative envelope of read–decide/refuse–execute–feedback/revision loops over a *witnessed* option set under a declared comparability relation; it has two layers (choosing within the structure / changing the structure).
Recoverable deprivation / live-field gap	Potential as a Readout [Lahtee, 2026i]; Choice [Lahtee, 2026j]	Uses a feasible counterfactual to ask how much potential or live field is recoverable; not automatically a "unit of severity"; diagnosing the compression is separated from identifying who is responsible (NC-79).
Power before choice	Choice Begins Before Choice [Lahtee, 2026j]	Synthesis: power can act on agency by changing meaning/accessibility before overt choice.
Human state $\rightarrow$ prompt	Fusion/Tunnel v8.1 [Lahtee, 2026f]; Meaning Before Naming [Lahtee, 2026g]; Experience v2 [Lahtee, 2026h]	The prompt is a bounded articulation of a history-bearing human state; Experience v2 names this the Pre-Prompt Human State Principle.

Table 1 continued

Segment in Master	Primary internal source (deposited record)	What that source actually supplies
Human ↔ AI recursion	Fusion/Tunnel v8.1 [Lahtee, 2026f]; Experience v2 [Lahtee, 2026h]	Each turn changes the context/readout of the next turn; AI output becomes a phenomenon that the human can give meaning to and retain.
Reciprocal momentum / amplification	Fusion/Tunnel v8.1 [Lahtee, 2026f]	Supplies the finite dialogue stepper, human/AI momentum, and χ_{recip} as a reciprocal-lineage diagnostic; amplification is not truth.
Session retention	Human LoRA [Lahtee, 2026c] → Fusion/Tunnel v8.1 [Lahtee, 2026f] (Repair 7)	v8.1 retains η_s only as a conceptual session-end retention gate and measures the residue that actually persists with the unaided return battery Y^{return} and RET, rather than inferring it from within-session engagement; the rank-restricted update formula in Section 3.11 is Master's own proposal (inheriting from an undeposited conversation-centered draft †).
Candidate status	When AI Expands Human Potential [Lahtee, 2026a]; Fusion/Tunnel v8.1 [Lahtee, 2026f]	AI is a candidate generator, not a certifier; v8.1 formalizes this as $\mathrm{AI}(Q) = K_{\text{like}} \neq K_{\text{validated}}$.
Difference–Resistance–Agency	Fusion/Tunnel v8.1 [Lahtee, 2026f]; normative ancestry from [Lahtee, 2026a]	D-R-A are constitutive conditions under the definition of durable human epistemic expansion; not an empirical universal law.
Expansion vs. tunnel direction	Fusion/Tunnel v8.1 [Lahtee, 2026f]	Separates reciprocal amplification, human retention, and epistemic direction into three axes; three-axis outcome $J_s^* = (\mathrm{AUG}_s, \mathrm{SYN}_s, \mathrm{RET}_s)$.
Human Return / CTSA	Fusion/Tunnel v8.1 [Lahtee, 2026f]; CTSA v9.1 [Lahtee, 2026k]	Fusion supplies the Human Return horizon; CTSA separates what returns into Concept, Tool, Skill, and Alternative, and audits Loss, Regulation, Provenance, Warrant, Direction.
New in v1.3 (three companion papers, 6 September 2026)		
Topic entry before the pre-prompt transport	How Humans Should Converse with AI (Dialogue Conversion Protocol) v1.3 [Lahtee, 2026o]	Places a Topic Entry Condition (LiveProblem/OpenExploration/RoutineDelegation) in front of the $H_{s,0} \to Q_{s,0}$ transport already in this river as (27); Problem-First is a default for human-development use, not a requirement (Problem-First ≠ Problem-Only).
Human Return closed through action and world feedback	Dialogue Conversion Protocol v1.3 [Lahtee, 2026o]	Extends the terminal Human Return audit with a further step: the human's integration record and selected action, and the world's returned record, feed into the next session's pre-prompt state, closing a Live-Problem → Revision-or-New-Problem cycle.
World-system human position (labour centrality, citizen claim, ownership, social reproduction, power channels)	After Labour v1.2 [Lahtee, 2026m]	Places corrigible agency, live possibility, and Human Return (already in this river, Sections 3.7 and 3.12) inside a wider accounting architecture of effective material claim, four-dimensional labour centrality, the Citizen Claim Threshold, ownership accumulation, and a typed Human Systemic Position audit index; non-collapse: aggregate output rising does not entail the audit index rising.
Human Conversion Vector and Reversibility Window	The Human Conversion Imperative v1.1 [Lahtee, 2026n]	Reframes machine-capability growth versus human conversion as a vector-valued, multi-dimensional elasticity problem rather than one augmentation score, and adds a Reversibility Window / Urgency Vector for when lost human capacities or routes become costly to restore.
New in v1.4 (one companion paper, 6 September 2026)		
Barrier readout before route generation	From Assistance to Human Capability (RG-HCA) v1.1 [Lahtee, 2026p]	Places a typed barrier ledger (Knowledge, Skill, Language, Tool, ResourceTime, Network, Credential, Permission, Opportunity, Unknown) between a human-owned live problem and any candidate route, guarding Observed Difficulty ≠ Skill Deficit (RG-HCA eq. 12–13).

Table 1 continued

Segment in Master	Primary internal source (deposited record)	What that source actually supplies
Endorsed capability/opportunity routes and scaffold fading	RG-HCA v1.1 [Lahtee, 2026p]	Separates AI-generated candidate routes from human-endorsed live routes (RG-HCA eq. 16), guards three non-collapse laws on proactive suggestion, capability advancement, and scaffolding (RG-HCA eq. 17–19), and gives a monotone OPEN fading rule tying assistance withdrawal to demonstrated Stable Unaided Return (RG-HCA eq. 22).
Opportunity conversion after Human Return	RG-HCA v1.1 [Lahtee, 2026p]	Extends the Human Return $\rightarrow$ Action $\rightarrow$ World Feedback closure already in this river (Section 3.14) with a world-side opportunity readout and the non-collapses Credential $\neq$ Capability, Market Legibility $\neq$ Human Worth (RG-HCA eq. 27–30), plus a net advancement record and a conditional estimand for policy comparison (RG-HCA eq. 36–37).

3 The Equation River According to Provenance

3.1 Foundation: Retained Difference, Readout, Meaning, Experience

The foundation of the programme begins from an architectural prohibition: a semantic noun is not a root primitive in the human domain. We therefore begin from the phenomenon / world-side encounter x_n and give it a finite human readout

$$r_n = R_H(x_n \mid H_n, c_n). \tag{1}$$

[Lahtee, 2026d,h]

Meaning-giving is written as a significance relation under human state, context, and criterion/question,

$$\mu_n = \Psi_H(r_n, H_n, c_n, Q_n), \tag{2}$$

and in Experience v2 the analytic modes can be separated as

$$\mu_n = (\mu_n^{\mathrm{aff}}, \mu_n^{\mathrm{prag}}, \mu_n^{\mathrm{auto}}, \mu_n^{\mathrm{conc}}, \mu_n^{\mathrm{epi}}). \tag{3}$$

[Lahtee, 2026h]

Readout Genesis (MEMK domain, eq. 14 of the deposited record) gives the experience equation as $E_{i,n} = \Phi_E(x_{i,n}, \mu_{i,n}, \kappa_{i,n}, c_n)$; dropping the reader index i and renaming κ_n to γ_n^{μ} to avoid a symbol collision (Section 4.1) gives

$$E_n = \Phi_E(x_n, \mu_n, \gamma_n^{\mu}, c_n), \tag{4}$$

where γ_n^{μ} is the strength/control of meaning over the current experience, so the defended compression is

$$\text{Experience} = \text{phenomenon-as-meaningfully-read}. \tag{5}$$

[Lahtee, 2026d,h]

3.2 Naming as Both an Articulation and a Restructuring Operator

Meaning Before Naming shows that explicit lexical naming need not precede experiential significance,

$$\ell_n = L_H(E_n, \mu_n \mid H_n, c_n), \qquad E_n, \mu_n \text{ may precede stable } \ell_n. \tag{6}$$

Experience v2 hardens this point by making naming recursive,

$$\mu_n \to E_n \to \ell_n \to \mu_{n+1} \to E_{n+1}. \tag{7}$$

[Lahtee, 2026g,h]

3.3 Retention and the Changing Future Reader

Human LoRA separates experience from durable adaptation in the master dependency form,

$$H_{n+1} = U_H\big(H_n,\ \mathrm{Retain}(E_n, \mu_n, r_n),\ \delta_{n:n+1}^{\mathrm{world}}\big). \tag{8}$$

[Lahtee, 2026c,h]

The important consequence is that retained experience can change the operator used to produce future readout; so the same biological individual may no longer be the same effective reader at a later time.

3.4 Resonance: Only the v2 Definition Is Current

The earlier Meaning Before Naming concept note used the word resonance close to an accessibility/reweighting diagnostic (and that document itself already flags a "literal-resonance error"); that definition has been superseded in Experience v2. The current canonical definition is external–internal experiential congruence, giving

$$I_{H,n} = \mathrm{Retrieve}(M_{H,n} \mid c_n, Q_n), \tag{9}$$

$$\mathrm{Reson}_H(n) = C_H(E_n^{\mathrm{cur}}, I_{H,n} \mid c_n, Q_n). \tag{10}$$

[Lahtee, 2026h]

Non-collapse must be preserved,

$$\begin{aligned} &\mathrm{Reson} \neq \mathrm{Identity}, \quad \mathrm{Reson} \neq \mathrm{Truth}, \\ &\mathrm{Reson} \neq \mathrm{Retention}, \quad \mathrm{Reson} \neq \mathrm{Improvement}. \end{aligned} \tag{11}$$

3.5 Rhythm, Momentum, and Accessibility Must Remain Parallel Descriptors

Rhythm is the temporal organization of the encounter history,

$$T_n = \{(x_k, t_k, w_k)\}_{k \le n}, \qquad \mathrm{Rhythm}_n = \Omega(T_n, B_n). \tag{12}$$

[Lahtee, 2026h]

From Problem to Hypothesis gives recent-path momentum as a separate quantity,

$$m_{t+1}(e) = \rho\, m_t(e) + \mathbb{1}[e_t = e], \qquad 0 \le \rho < 1, \tag{13}$$

and history-shaped semantic accessibility,

$$\begin{aligned} \kappa_{t+1}^{\mathrm{sem}}(e \mid Q) \ \propto\ \kappa_t^{\mathrm{sem},(0)}(e \mid Q) \exp\big(&\beta a_{Q,t}(e) + \mu m_t(e) \\ &- \nu c_{A,t,Q}(e) + \xi_t(e)\big). \end{aligned} \tag{14}$$

[Lahtee, 2026e]

The canonical correction is not to write Rhythm $\Rightarrow m_t$ as a derivation: rhythm, recent-path momentum, attraction, and switching cost are separate history/control descriptors that may jointly determine accessibility,

$$\mathrm{Rhythm}_t,\ m_t,\ a_t,\ c_t \longrightarrow \kappa_{t+1}^{\mathrm{sem}}. \tag{15}$$

3.6 Accumulation, Barrier, and Release

The MEMK domain in Readout Genesis has a semantic potential V_{sem}, informational work W_{info}, and activation barrier $\Delta V_{\mathrm{old}\to\mathrm{new}}^{\dagger}$, explicitly stated not to be physical energy. Experience v2 uses this to interpret retained transition readiness,

$$\Delta W_{j,N} = \sum_{n=1}^{N} \eta_n\, \mathrm{eligibleGradient}_n, \tag{16}$$

$$W_n^{\mathrm{eff}} = W_{\mathrm{info},n}\, g\big(\mathrm{Reson}_H(n),\ \mathrm{eligibility},\ \mathrm{context}\big), \tag{17}$$

and the transition candidate,

$$\sum_{k \le n} W_k^{\mathrm{eff}} \ \ge\ \Delta V_{\mathrm{old}\to\mathrm{new}}^{\dagger}. \tag{18}$$

[Lahtee, 2026d,h]

This is a candidate transition topology, not a claim that every kind of human change obeys a universal threshold. Release is not equal to transformation; intensity is not equal to truth; barrier crossing is not equal to improvement.

3.7 From Accessibility to the Live Possibility Field

Choice Begins Before Choice fills the missing interface between feasible capacity and overt choice,

$$\Pi^{\mathrm{live}}_{A,t}(g) \subseteq \Pi^{\mathrm{feas}}_{A,t}(g) \subseteq \Pi^{\mathrm{phys}}_{t}(g). \tag{19}$$

Let $\lambda^{\mathrm{live}}_{A,t}(\pi \mid g)$ be the practical accessibility weight of a feasible policy (renamed from the local $\kappa_{A,t}$ so as not to collide with semantic accessibility),

$$L_{A,t}(g) = \{(\pi, \lambda^{\mathrm{live}}_{A,t}(\pi \mid g)) :\ \pi \in \Pi^{\mathrm{feas}}_{A,t}(g)\}, \tag{20}$$

and the operational live set,

$$\Pi^{\mathrm{live}}_{A,t}(g) = \{\pi :\ \lambda^{\mathrm{live}}_{A,t}(\pi \mid g) \geq \tau_{\mathrm{live}}\}. \tag{21}$$

So overt choice is a downstream object,

$$\pi^{\mathrm{choice}}_{A,t} \in \Pi^{\mathrm{live}}_{A,t}(g), \tag{22}$$

but enactment may differ from choice, and observation is not the whole trajectory,

$$\pi^{\mathrm{act}} \neq \pi^{\mathrm{choice}} \text{ possible}, \qquad Y_{\mathrm{obs}} = \mathscr{O}_q(H_{0:T}), \qquad Y_{\mathrm{obs}} \neq H_{0:T}. \tag{23}$$

This is the non-collapse chain,

$$\text{possible} \neq \text{feasible} \neq \text{live} \neq \text{chosen} \neq \text{enacted} \neq \text{observed}. \tag{24}$$

[Lahtee, 2026j]

3.8 Potential as a Readout: The Two-Layer Witnessed Envelope

The note *Effective, Corrigible Agency Potential* (5 September 2026, not separately deposited) gave four joint conditions for the agency loop and has since been replaced by *Potential as a Readout* v2, which changes the set over which the maximum is taken from a declared feasible set to a **witnessed set**: the finite set of policies with a record that a comparable agent (under a declared comparability relation) actually used within horizon T and budget B. The canonical form in Master is therefore

$$p^{*}_{A,g}(h,z;T,B,P) = \max_{\pi \in \Pi^{\mathrm{wit}}_{A}(g;h,z,T,B)} \Pr^{\pi}_{P}\left(\mathrm{Read}_g \cap D_g \cap X_g \cap F_g\right). \tag{25}$$

[Lahtee, 2026i]

with a second layer $p^{*(2)}_{A,g} = \max_{z \in J_{\mathrm{feas}}} p^{*}_{A,g}(h,z;\cdot)$ (changing the structure z itself), and the recoverable gap being the difference between the two layers. Choice Begins Before Choice does not replace this equation but adds the pre-choice restriction that the currently considered policy must come from $\Pi^{\mathrm{live}} \subseteq \Pi^{\mathrm{feas}}$; so the envelope asks "what has actually been usable under the given resources/horizon", while the live field asks "what is ready as a candidate for this person right now".

Corrected from v1.0: v1.0's equation (25) used $\Pi^{\mathrm{feas}}_{A}(g)$, following the note that has since been replaced; the central claim of *Potential as a Readout* is that potential is a readout over the set that has witnesses only, and a supremum over an alternative with no record is not a readout (NC-78).

3.9 Power Before Choice, and the Recoverable Live-Field Gap

The Choice paper sharpens the political proposal: power need not wait until an already-perceived option is prohibited; it may change intelligibility, salience, social permission, expected consequence, and practical accessibility before overt choice. The recoverable live-field gap is written schematically as

$$L^{\mathrm{live}}_{A,g} = \max_{z \in J_{\mathrm{feas}}} D_L\left(L^{z}_{A}(g),\ L^{z_0}_{A}(g)\right). \tag{26}$$

[Lahtee, 2026j]

But L^{live} is not a violence score. Calling it structural deprivation/violence still requires a feasible counterfactual and an independent normative loss; diagnosing the narrowing of space (from an obstruction account and asymmetry of cost) is a different question from identifying who is responsible, which requires a chooser of the regime (NC-79).

3.10 Human–AI Coupling Begins Before the Prompt

Fusion/Tunnel and Meaning Before Naming rest on the premise that language is a bounded transport of human state. Experience v2 names this explicitly as the Pre-Prompt Human State Principle,

$$H_t \xrightarrow{L_H} Q_t, \tag{27}$$

where the AI receives Q_t, not the whole of H_t. The output returns as a human encounter,

$$Q_t \to \mathrm{AI}_t \to Y_t \xrightarrow{R_H} E^{\mathrm{AI}}_{H,t}, \tag{28}$$

$$H_{t+1} = U_H\left(H_t, E^{\mathrm{AI}}_{H,t}, \delta^{\mathrm{world}}, X^{\mathrm{other}}\right), \tag{29}$$

and if a residue persists, it is then possible that

$$L_{A,t+1} \neq L_{A,t} \tag{30}$$

before the next prompt is generated.

[Lahtee, 2026f,h,j]

3.11 Within-Session Amplification, Retention, and Epistemic Direction

Fusion/Tunnel v8.1 gives the finite dialogue state and reciprocal-lineage diagnostic,

$$Z_{\mathrm{dlg}}[s,n+1] = F^{\#}_{\mathrm{dlg}}\left(Z_{\mathrm{dlg}}[s,n], u_H[s,n], u_{\mathrm{AI}}[s,n], c[s,n], T[s,n]\right), \tag{31}$$

$$\chi_{\mathrm{recip}}[s,n,L] = \frac{|D_{\mathrm{recip}}[s,n,L]|}{|\Sigma[s,n]|}. \tag{32}$$

[Lahtee, 2026f]

χ_{recip} measures the finite reciprocal-descendant gain before epistemic pruning; it is not warrant or a truth score.

Session retention. v8.1 (Repair 7) states that "a scalar η_s should not be inferred from in-session engagement or recall alone": it keeps η_s as a conceptual session-end retention gate but estimates the residue that actually persists through an unaided return battery,

$$Y^{\mathrm{return}}_{s+\Delta} = (R_{\mathrm{rec}}, R_{\mathrm{disc}}, T_{\mathrm{new}}, Q_{\mathrm{next}}), \tag{33}$$

and, for randomized study designs, a difference-in-differences estimand,

$$\mathrm{RET} = \left(P^{\mathrm{post}}_{H,\mathrm{AI}} - P^{\mathrm{pre}}_{H,\mathrm{AI}}\right) - \left(P^{\mathrm{post}}_{H,C} - P^{\mathrm{pre}}_{H,C}\right). \tag{34}$$

[Lahtee, 2026f]

Master's own notation proposal for writing retention as a restricted update (inheriting from Fusion's undeposited conversation-centered draft v3, which doubles η_s — Section 4.3) is to separate the candidate restricted update from the retention gate,

$$\widetilde{\Delta\Omega}_s^H = B_s A_s, \qquad \operatorname{rank}(B_s A_s) \ll d_H, \tag{35}$$

$$\Omega_{s+1,0}^H = \Omega_{s,0}^H + \eta_s \widetilde{\Delta\Omega}_s^H, \qquad 0 \le \eta_s \le 1. \tag{36}$$

Equations (35)–(36) are *Master's own proposal*, preserving the intent of Human LoRA/Fusion that exposure is not equal to retention, without producing an η_s^2 term; they are not v8.1's own text, and v8.1's (33)–(34) remain the measurement actually tied to Human Return.

Within the scope of Fusion/Tunnel, AI-side semantic momentum is reset across sessions,

$$m_{s+1,0}^{\mathrm{AI}} = 0, \tag{37}$$

but this line is a scope condition of the model, not a universal fact about AI with persistent memory.

Candidate status and direction are kept separate from fluency/amplification,

$$\mathrm{AI}(Q) = K_{\mathrm{like}}, \qquad K_{\mathrm{like}} \ne K_{\mathrm{validated}}, \tag{38}$$

$$\Delta_s := G_s - T_s, \qquad \operatorname{sign}(\Delta P_H(s)) = \operatorname{sign}(\eta_s \Delta_s), \qquad \eta_s \ge 0, \tag{39}$$

and the three-axis outcome that v8.1 forbids collapsing into a single score,

$$\begin{aligned} J_s^* &= (\mathrm{AUG}_s,\ \mathrm{SYN}_s,\ \mathrm{RET}_s), \\ \mathrm{AUG}_s &= P_s^{\mathrm{joint}} - P_s^H, \\ \mathrm{SYN}_s &= P_s^{\mathrm{joint}} - \max(P_s^H, P_s^{\mathrm{AI}}). \end{aligned} \tag{40}$$

[Lahtee, 2026a,f]

3.12 Human Return and CTSA as the Upper Measurement Boundary

The Human Return horizon asks what, once AI is removed, can still be reconstructed, discriminated, transferred, and generated by the person alone. CTSA v9.1 separates retained capability into Conceptual, Tool-Selection, Skill-Execution, and Alternative-Generation Return with an audit record, originally written as R_H; Master renames it H_{return} so as not to collide with R's other roles (Section 4.4),

$$H_{\mathrm{return}} = \langle G_{\mathrm{CTSA}}, L, M, P, W, \Delta_{\mathrm{dir}} \rangle, \tag{41}$$

where G_{CTSA} = retained gains, L = losses, M = metacognitive regulation, P = provenance, W = warrant, Δ_{dir} = epistemic direction.

[Lahtee, 2026k]

The end of the chain therefore preserves an important non-collapse,

$$\begin{aligned} &\text{Exposure} \ne \text{Retention} \ne \text{Improvement}, \\ &\text{Accessibility} \ne \text{Warrant} \ne \text{Truth}. \end{aligned} \tag{42}$$

3.13 Topic Entry and Agenda Ownership Before the Prompt

Equation numbers in this river do not run in strict page order: Sections 3.13 to 3.15 introduce (46)–(64) at this point in the text, ahead of (43)–(45), which keep the section numbers already assigned to them in v1.1/v1.2 (Sections 4, 5 and 8) and so print later, on the following pages. A citation such as "eq. (45)" or "eq. (64)" should therefore be located by its cross-reference label or by section number, not by page order.

The pre-prompt transport already given in this river as (27), $H_t \xrightarrow{L_H} Q_t$, assumes that some topic has already captured the human's attention. The Dialogue Conversion Protocol (DCP) places a Topic Entry Condition in front of this transport, restated here with DCP's own session index s [Lahtee, 2026o], eq. (1):

$$H_{s,0} \xrightarrow{L_H} Q_{s,0}. \tag{46}$$

For human-development use, DCP defines a live problem as a presently consequential unresolved difference in the person's life, work, understanding, relationship to evidence, or environment (DCP eq. 4), and separates three legitimate entry modes (DCP eq. 5):

$$\begin{aligned} \text{TopicEntry} \in \{&\text{LiveProblem},\ \text{OpenExploration}, \\ &\text{RoutineDelegation}\}. \end{aligned} \tag{47}$$

LiveProblem is the default when the goal is learning, judgment, problem solving, or human development; OpenExploration protects curiosity, play, art, and basic inquiry; RoutineDelegation covers low-value repetitive tasks. DCP's Problem-First Dialogue Principle [OPEN] states a fourfold rationale (DCP eq. 6):

$$\text{Problem-First} \Rightarrow \begin{cases} \text{Human Agenda Ownership}, \\ \text{Bounded Relevance}, \\ \text{World-Side Testability}, \\ \text{Observable Human Return}. \end{cases} \tag{48}$$

Problem-first is a default, not a monopoly; DCP is explicit about the guarding non-collapse (DCP eq. 7, restated at DCP eq. 43):

$$\text{Problem-First} \ne \text{Problem-Only}. \tag{49}$$

[Lahtee, 2026o]

The transport (27)/(46) into this river should therefore be read as occurring downstream of a topic-entry decision rather than as beginning from an already-given topic; the governing question is who selected the topic and whether the human still recognizes it as their own, not merely whether the topic is currently useful.

3.14 Human Return, Action, and World Feedback

Once Human Return (41)–(42) has been read, DCP does not treat verbal reconstruction as the end of a human-development conversation when the original topic was action-relevant. Let $I_s = \langle \Delta M_s, E_s^{\mathrm{decisive}}, U_s^{\mathrm{remain}}, \mathrm{Next}_s \rangle$ be DCP's integration record (DCP eq. 22), a_s the human-selected next action, and $\delta_{s+1}^{\mathrm{world}}$ the returned record from the situation that generated the problem. DCP's closure (DCP eq. 29) is

$$I_s \to a_s \to \delta_{s+1}^{\mathrm{world}} \to H_{s+1,0}. \tag{50}$$

The wider cycle DCP proposes (DCP eq. 30) is

$$\begin{aligned}\text{Live Problem} &\to \text{Question} \to \text{Dialogue}\\ &\to \text{Human Return} \to \text{Action}\\ &\to \text{World Feedback} \to \text{Revision or New Problem.}\end{aligned} \tag{51}$$

[Lahtee, 2026o]

This is conditional, not universal: DCP states that pure learning, creative exploration, and basic inquiry can terminate at a different boundary, and its World-closure proposition [OPEN] applies only where the original topic had observable consequences. Within this river, (50)–(51) extend the Human Return boundary of Section 3.12 with a further arrow, $H_{\text{return}} \to a_s \to \delta^{\text{world}} \to H_{s+1,0}$, which Section 5 appends to the corrected river as an outer-loop tail.

3.15 The World-System Layer: Human Systemic Position and Human Conversion

After Labour and The Human Conversion Imperative place the corrigible-agency, live-possibility, and Human Return objects of Sections 3.7, 3.8 and 3.12 inside a wider world-system accounting layer, opening with a self-contained readout definition (After Labour eq. 2):

$$\text{Readout}_{Q,O,c}(S) = z, \qquad z \neq S. \tag{52}$$

[Lahtee, 2026m]

Labour's systemic role is disaggregated into four coordinates rather than read from employment alone (After Labour eq. 8):

$$L_t = \langle L_t^{\text{task}}, L_t^{\text{income}}, L_t^{\text{bottleneck}}, L_t^{\text{bargain}} \rangle, \tag{53}$$

and the minimum non-labour conversion needed to hold a target material claim $\bar{\Gamma}$ against a falling labour-income share s_t^L is the Citizen Claim Threshold, an accounting identity inside the stylized claim block (After Labour eq. 13):

$$q_t^{\min} = \frac{\bar{\Gamma} - s_t^L}{1 - s_t^L}. \tag{54}$$

After Labour restates, for a person i and goal g, the same possible/feasible/live nesting already used in Section 3.7 (After Labour eq. 32, reusing [Lahtee, 2026j]):

$$\Pi_{i,t}^{\text{live}}(g) \subseteq \Pi_{i,t}^{\text{feas}}(g) \subseteq \Pi_{i,t}^{\text{phys}}(g), \tag{55}$$

and a task-relative corrigible-agency envelope over that live set (After Labour eq. 34):

$$A_{H,i,t}^{\text{corr}}(g) = \max_{\pi \in \Pi_{i,t}^{\text{live}}(g)} \Pr^{\pi}(R_g \cap D_g \cap X_g \cap F_g), \tag{56}$$

the same task-relative envelope form as (25), restated here at the world-system level rather than the witnessed-set level. A delayed unaided return profile is likewise restated at this level (After Labour eq. 35; its four coordinates parallel H_{return} in (41) but are kept under a different symbol, since After Labour does not assert the two instruments are identical):

$$R_{H,t}^{\text{return}} = \langle C_t, T_t, S_t^{\text{skill}}, A_t^{\text{alt}} \rangle. \tag{57}$$

These feed a typed audit index, the Human Systemic Position (After Labour eq. 40):

$$P_t^H = \frac{(\Gamma_t^{\text{eff}})^{\theta_\Gamma} (A_{H,t}^{\text{corr}})^{\theta_A} (\Lambda_{H,t}^{\text{live}})^{\theta_\Lambda} (r_t^H)^{\theta_R} (S_t^H)^{\theta_S} (X_t^H)^{\theta_X}}{(1 + D_t^H)^{\theta_D}}, \tag{58}$$

which After Labour states explicitly is not a welfare utility function or a validated cardinal scale but a typed audit index, guarding against output, consumption, or transfer income silently standing in for human position. Its primary non-collapse result (After Labour eq. 43) is

$$\dot{Y}_t > 0 \not\Rightarrow \dot{P}_t^H > 0. \tag{59}$$

[Lahtee, 2026m]

The Human Conversion Imperative generalizes the same worry into one framing inequality (HCI eq. 1):

$$\text{Machine expansion} \not\Rightarrow \text{Human expansion}, \tag{60}$$

and replaces a scalar augmentation score with a Human Conversion Vector [DEFINITION] (HCI eq. 11):

$$\begin{aligned} C_t^H = \big\langle \Gamma_t^{\text{eff}}, X_t^H, \Lambda_{H,t}^{\text{live}}, A_{H,t}^{\text{corr}}, R_{H,t}^{\text{route}}, \\ W_{H,t}^{\text{world}}, r_{H,t}, H_t^{\text{cap}}, S_{H,t} \big\rangle, \end{aligned} \tag{61}$$

with a local elasticity for each dimension j against a declared machine-capability measure M_t [DEFINITION] (HCI eq. 12):

$$\eta_{j,t}^{HC} = \frac{\partial \ln C_{j,t}^H}{\partial \ln M_t}, \tag{62}$$

which permits mixed trajectories in which material claim rises while retained competence or exit fall. Because adoption can display increasing returns and path dependence, HCI defines a Reversibility Window for each dimension j [DEFINITION] (HCI eq. 17):

$$W_j = \{ t : C_{j,t}^{\text{rec}} \leq \bar{C}_j \wedge \tau_{j,t}^{\text{rec}} \leq \bar{\tau}_j \}, \tag{63}$$

and an Urgency term combining the machine/human growth gap with lock-in, severity, and recovery time [DEFINITION] (HCI eq. 19):

$$U_{j,t} = \Delta g_{j,t} L_{j,t} S_{j,t} \tau_{j,t}^{\text{rec}}. \tag{64}$$

[Lahtee, 2026n]

HCI's Reversibility Principle (its Proposition 3) is explicitly OPEN: the faster machine capability grows relative to human conversion, the more valuable it becomes to preserve reversible human and institutional alternatives before the recovery cost of lost routes rises. Neither source paper claims that P_t^H, C_t^H, $\eta_{j,t}^{HC}$, W_j, or $U_{j,t}$ is a validated causal or welfare measure; both state that weights, elasticities, and thresholds must be declared and estimated per study, and that the architecture should be narrowed or removed if it fails to discriminate outcomes beyond ordinary productivity, wage, or skill measures.

3.16 Barrier Readout, Endorsement, Scaffold Fading, and Opportunity Conversion

From Assistance to Human Capability (RG-HCA) proposes a proactive extension of the Dialogue Conversion Protocol tail already in this river (Sections 3.13 and 3.14): once a live problem is human-owned, AI need not remain purely reactive. RG-HCA is explicit that this is a candidate domain translation, not a closed human science, and keeps the constitutional non-collapse $S_n \neq Z_{\mathrm{HCA},n} \neq D_{\mathrm{HCA},n}$ (RG-HCA eq. 4). Its own native river (RG-HCA eq. 5) is

$$\begin{aligned} \text{Retained Difference} &\to \text{Human Readout} \\ &\to \text{Live Problem} \\ &\to \text{Barrier Readout} \\ &\to \text{Candidate Routes} \\ &\to \text{Human Endorsement} \\ &\to \text{Adaptive Scaffold} \\ &\to \text{Practice} \\ &\to \text{Withdrawal} \\ &\to \text{Human Return} \\ &\to \text{Novel Transfer}^{1} \\ &\to \text{World Feedback} \\ &\to \text{Opportunity Conversion.} \end{aligned} \tag{67}$$

[Lahtee, 2026p]

Read against (51), (67) inserts a Barrier Readout stage before Candidate Routes and appends Opportunity Conversion after World Feedback; it is a proactive elaboration of the same Live-Problem-to-Revision cycle, not a replacement of it.

Two agents given the same AI do not meet on a blank plane. RG-HCA bookkeeps this as a Life-Capital Context Vector [Definition] (RG-HCA eq. 8):

$$\begin{aligned} K^{\mathrm{life}}_{i,n} = \big\langle & E^{\mathrm{econ}}, F^{\mathrm{base}}, L^{\mathrm{lang}}, D^{\mathrm{digital}}, T^{\mathrm{disc}}, \\ & H^{\mathrm{health}}, M^{\mathrm{mob}}, N^{\mathrm{mentor}}, C^{\mathrm{cred}} \big\rangle_{i,n}, \end{aligned} \tag{68}$$

[Lahtee, 2026p]

denoting economic resources, foundational literacy/numeracy access, language bridge, device/connectivity, discretionary time, health/nutrition/sleep conditions, mobility/safety, mentor/network access, and credential pathway; RG-HCA states this is not a universal cardinal measure of human capital and does not exhaust life conditions.

The proactive step begins only after the problem is human-owned, and inserts a barrier-typing stage before any route or training prescription. RG-HCA's typed barrier ledger (RG-HCA eq. 12) is

$$\begin{aligned} B^{\mathrm{bar}}_{i,n} \subseteq \{ & \mathrm{Knowledge}, \mathrm{Skill}, \mathrm{Language}, \mathrm{Tool}, \\ & \mathrm{ResourceTime}, \mathrm{Network}, \mathrm{Credential}, \\ & \mathrm{Permission}, \mathrm{Opportunity}, \mathrm{Unknown} \}, \end{aligned} \tag{69}$$

[Lahtee, 2026p]

and its load-bearing non-collapse [Definition] (RG-HCA eq. 13) is

$$\text{Observed Difficulty} \neq \text{Skill Deficit.} \tag{70}$$

[Lahtee, 2026p]

RG-HCA gives concrete costs of a wrong diagnosis: repeated skill drills punish a translation problem when the barrier is language, and more private competence stays invisible to an employer when the barrier is a missing credential.

RG-HCA separates candidate routes from endorsed routes so that proactive suggestion never silently becomes the person's own goal. The endorsed subset [Definition] (RG-HCA eq. 16) is

$$C^{\mathrm{live}}_{i,n} = \{ c \in C^{\mathrm{cand}}_{i,n} : \ \mathrm{Endorse}_i(c) = 1 \}, \tag{71}$$

[Lahtee, 2026p]

and three further non-collapse laws follow [Definition] (RG-HCA eq. 17–19):

$$\begin{gathered} \text{Proactive Suggestion} \neq \text{Human Goal Ownership,} \\ \text{Capability Advancement} \neq \text{AI Goal Authority,} \\ \text{Scaffolding} \neq \text{Control.} \end{gathered} \tag{72}$$

[Lahtee, 2026p]

Once a route is endorsed and scaffolded, RG-HCA does not prescribe a fixed amount of assistance. Its candidate fading rule [OPEN] (RG-HCA eq. 22) is

$$\text{Stable Unaided Return} \uparrow \Rightarrow \ h^{\mathrm{decisive}} \downarrow, \tag{73}$$

[Lahtee, 2026p]

unless task difficulty, stakes, or domain risk rises; RG-HCA marks this rule Open and states it should be replaced if a simpler rule performs as well.

RG-HCA's own Human Return reuse (its eq. 24–25) restates $R^{\mathrm{return}}_{H} = \langle C, T, S, A \rangle$ and the non-collapse $\Delta\mathrm{Performance}_{\mathrm{AI}} > 0 \not\Rightarrow \Delta R^{\mathrm{return}}_{H} > 0$. RG-HCA cites the same CTSA v9.1 Human Return source already cited at this river's H_{return} (41) and its non-collapse (42) (Section 3.12), but decomposes it as a four-field tuple $\langle C, T, S, A \rangle$ with a differently-worded non-collapse, rather than the six-field $\langle G_{\mathrm{CTSA}}, L, M, P, W, \Delta_{\mathrm{dir}} \rangle$ of (41). The underlying evidentiary standard – unaided reconstruction, selection, execution/transfer, or generation under a declared criterion – is the same shared lineage; the two pairs of equations are not identical, so RG-HCA's eq. 24–25 are described here in prose rather than given separate boxed equation numbers.

After Human Return, RG-HCA does not treat verbal reconstruction as closing an action-relevant problem, restating the same world-closure step as (50) over RG-HCA's own candidate HCA state Z_{HCA} (RG-HCA eq. 27):

$$R^{\mathrm{return}}_{H,i,n} \ \to \ a_{i,n} \ \to \ \delta^{\mathrm{world}}_{i,n+1} \ \to \ Z_{\mathrm{HCA},i,n+1}. \tag{74}$$

[Lahtee, 2026p]

A further boundary appears after Human Return: the person may now be capable yet still lack a viable social or economic route. RG-HCA defines a world-side opportunity readout (RG-HCA eq. 28)

$$\begin{aligned} \Omega^{\mathrm{real}}_{i,n} = G_O\big(& R^{\mathrm{return}}_{H,i,n}, K^{\mathrm{life}}_{i,n}, \\ & \mathrm{Cred}_{i,n}, \mathrm{Net}_{i,n}, \mathrm{Perm}_{i,n}, \\ & \mathrm{MarketReadout}_n \big), \end{aligned} \tag{75}$$

[Lahtee, 2026p]

[1] Stage label quoted verbatim from RG-HCA's own river (its eq. 5), as in v1.4; it names a stage of that source's pipeline and is not a claim of this document about priority.

explicitly not a validated welfare utility function, and two further non-collapses are essential [Definition] (RG-HCA eq. 29–30):

$$\begin{aligned} \text{Credential} &\neq \text{Capability}, \\ \text{Market Legibility} &\neq \text{Human Worth}. \end{aligned} \tag{76}$$

[Lahtee, 2026p]

RG-HCA uses Spence's signalling model only to motivate this legibility gap: capability that is not represented by a recognizable credential, portfolio, demonstration, reference, or network may fail to become opportunity, so the remedy can be an opportunity bridge rather than more skill training.

A full evaluation must separate immediate assisted performance, Human Return, transfer, autonomy, burden, and opportunity conversion rather than collapsing them into current task success. RG-HCA's net advancement record [Definition] (RG-HCA eq. 36) is

$$\begin{aligned} A_i^{\mathrm{HCA}} = \langle &\mathrm{Gain}_{CTSA}, \mathrm{Loss}, \mathrm{Transfer}, \\ &\mathrm{Ownership}, \mathrm{Burden}, \mathrm{BarrierChange}, \\ &\mathrm{OpportunityChange}, \mathrm{Provenance}, \mathrm{Warrant}\rangle, \end{aligned} \tag{77}$$

[Lahtee, 2026p]

and for policy comparison under unequal starting conditions RG-HCA replaces a raw outcome contrast with a conditional estimand [Definition, measurement] (RG-HCA eq. 37):

$$\mathrm{ATE}_{\mathrm{HCA}}(x) = \mathbb{E}\big[Y(1) - Y(0) \mid K^{\mathrm{life}} = x\big], \tag{78}$$

[Lahtee, 2026p]

where x denotes a declared baseline life-context stratum; RG-HCA notes that matching or covariate adjustment is a design strategy, not part of the estimand itself.

Human Capability Advancement Principle [Open], restated from RG-HCA. When durable human development is an endorsed goal, AI may proactively identify and propose a reachable capability or opportunity expansion, but it should remain subordinate to human-owned ends, prefer the least burdensome intervention likely to produce durable return, and withdraw decisive support as the person demonstrates independent capability [Lahtee, 2026p].

None of (67)–(78) replaces or reweights (44)–(51); RG-HCA is read here as a proactive layer that a completed river may attach at the human-owned-problem and Human-Return boundaries already fixed in Sections 3.12 to 3.14, carrying its own Open status where RG-HCA itself marks it Open and no stronger tier than [Definition] where RG-HCA marks a bookkeeping object.

4 Canonical Notation Corrections Before Freezing Master

The audit found five points that must be corrected from the local notation of each individual paper before combining them into the canonical programme equation.

4.1 κ Collides Across Three Meanings

In MEMK, κ_n means meaning strength; in From Problem to Hypothesis, $\kappa_t(e \mid Q)$ means semantic accessibility; in Choice Begins Before Choice, $\kappa_{A,t}(\pi \mid g)$ means the practical accessibility of a policy. Master should use one consistent rename,

$$\begin{aligned} \gamma_n^{\mu} &= \text{meaning strength}, \\ \kappa_{A,t}^{\mathrm{sem}}(e \mid Q) &= \text{semantic-route accessibility}, \\ \lambda_{A,t}^{\mathrm{live}}(\pi \mid g) &= \text{live-policy accessibility}. \end{aligned} \tag{43}$$

4.2 Rhythm Does Not Automatically Derive Momentum

Rhythm and momentum are parallel history descriptors. Master should therefore write $\mathrm{Rhythm}_t, m_t, a_t, c_t \to \kappa_{t+1}^{\mathrm{sem}}$ as a dependency/hypothesis interface, rather than writing Rhythm $\Rightarrow$ Momentum.

4.3 The η_s Double-Gating: From a Draft, Not From v8.1

Fusion's conversation-centered draft (v3, not deposited) writes $\Delta\Omega_s^H \approx \eta_s B_s A_s$ and then updates state with $+\eta_s \Delta\Omega_s^H$ again; read literally this produces η_s^2. The deposited record (v8.1) no longer contains this rank-update formula and instead chooses to measure return via (33)–(34). Master's proposal (35)–(36) is therefore a proposed canonical form for the programme, not a correction of v8.1's own text.

Corrected from v1.0: v1.0 attributed this formula to "Fusion/Tunnel v7" and called it a canonical correction of the source; an independent reviewer searched the v8.1 text and did not find this formula (searches for LoRA / rank / B_s / A_s / double $= 0$), so it has been reassigned the status above.

4.4 The Symbol R Collides Across Several Roles

The programme previously used R for the read condition, Resistance, Resonance, the Rhythm descriptor, and Human Return (R_H in CTSA). Master uses clearly typed names: Read, Resist, Reson, Rhythm, and H_{return} (renamed from CTSA's R_H; the content is identical).

4.5 $\Pi^{\mathrm{feas}} \to \Pi^{\mathrm{wit}}$ in the Potential Equation

Per *Potential as a Readout*, the set over which the maximum is taken must be a finite set with witnesses under a declared comparability relation (Section 3.8); Π^{feas} remains valid in Choice's sense (structurally feasible) but is not the domain of max in (25).

5 The Corrected Master Equation River

After fixing the symbol collisions and role separations, the canonical river that is most faithful to the internal literature is

$$
\begin{aligned}
x_t &\xrightarrow[\text{Genesis}]{R_H} r_t \xrightarrow[\text{Experience}]{\Psi_H} \mu_t \xrightarrow[\text{MEMK}]{\Phi_E} E_t \\
&\xrightarrow[\text{Human LoRA}]{\text{Retain}/U_H} H_{t+1} \\
&\xrightarrow[\text{Resonance, Rhythm, Attraction/Momentum}]{} \lambda^{\text{live}}_{A,t+1}(\pi \mid g) \\
&\to L_{A,t+1}(g) \to \Pi^{\text{live}}_{A,t+1}(g) \xrightarrow[\text{Choice Before Choice}]{} \pi^{\text{choice}} \\
&\xrightarrow[\text{Potential as a Readout}]{\text{Read},D,X,F} \pi^{\text{act}} \to \text{feedback/correction} \to H_{t+2} \\
&\xrightarrow[\text{Pre-Prompt State}]{L_H} Q_{t+2} \to \text{AI} \to K_{\text{like}} \\
&\xrightarrow[\text{Fusion/Tunnel}]{\text{Difference + Resistance, } \eta_s} \text{recursive dialogue} \\
&\to \text{retained human residue} \xrightarrow[\text{CTSA}]{\text{AI removed}} H_{\text{return}} \to \text{Epistemic Direction.}
\end{aligned} \tag{44}
$$

The important point is that equation (44) is not a claim that every arrow carries the same evidential status. Genesis/MEMK supply architectural order; resonance/rhythm/live-field are domain/Dr/Open constructs; Potential as a Readout is a task-relative measurement architecture (definitions at the definition level, lemmas on a proven discrete model, and empirical proposals that remain open); Fusion/Tunnel carries both constitutive and open diagnostics; CTSA is a session-boundary measurement interface.

Per the Dialogue Conversion Protocol's closure (Section 3.14), the chain in (44) continues past H_{return} rather than terminating there; this is an appended tail, not a rewriting of (44) itself, which still ends at Human Return / Epistemic Direction exactly as corrected above:

$$
\cdots \to H_{\text{return}} \to a_s \to \delta^{\text{world}} \to H_{t+3}. \tag{65}
$$

After Labour and the Human Conversion Imperative (Section 3.15) do not add a further arrow inside this cycle; they sit as an outer loop that reads the cycle's own state at each turn. Writing $t \to t+3$ for one pass of (44)–(65), the outer loop is schematically

$$
\begin{aligned}
&\left(H_t \xrightarrow[(44),(65)]{} H_{t+3}\right) \xrightarrow[\text{Readout}_{Q,O,c}]{} C^H_{t+3} \to P^H_{t+3} \\
&\qquad \xrightarrow[\Gamma^{\text{eff}},\Pi^{\text{live}},A^{\text{corr}},R^{\text{return}}]{} \left(H_{t+3} \xrightarrow[(44),(65)]{} H_{t+6}\right).
\end{aligned} \tag{66}
$$

Equation (66) is a schematic placement, not a new derivation: it records that After Labour's P^H_t and HCI's C^H_t are readouts of successive human-return cycles, and that the world-system coordinates they track (effective material claim, live possibility, corrigible agency, Human Return, social standing) condition the next cycle's live field and agency envelope in (44) without replacing anything inside it. The non-collapse of Section 3.15 applies at this outer scale too: a cycle that raises assisted performance or aggregate output does not by itself raise P^H_t ((59)).

RG-HCA's barrier and opportunity layer (Section 3.16) sits inside this same outer loop, between the Human-Return tail H_{t+3} of (65) and the world-system readout that opens (66), restating (74)–(75) at the scale of one full pass:

$$
H_{t+3} \xrightarrow[(69),(71)]{\text{Barrier, Endorsement}} Z_{\text{HCA},t+3} \xrightarrow[(75)]{\Omega^{\text{real}}} \left(C^H_{t+3} \to P^H_{t+3}\right). \tag{79}
$$

Equation (79) is likewise a schematic placement: it records that RG-HCA's barrier ledger, endorsement, and opportunity readout condition what of H_{t+3} becomes the world-system state C^H_{t+3} already read in (66), without adding a further arrow inside (44)–(65) or reweighting (66) itself.

6 The Evolution of Questions in the Corpus

1. **When AI Expands Human Potential** (March 2026) [Lahtee, 2026a]: asks when AI expands human potential, and answers normatively that it must add corrigibility, world-responsiveness, route-openness, and disciplined revision — not merely productivity or fluency.
2. **Human Learning as Epistemic Architecture** (June 2026) [Lahtee, 2026b]: shifts the centre to human learning: lived words, meaning, relation, retrieval, constraint, revision, and competence feedback.
3. **Experience Is the Human LoRA** (July 2026) [Lahtee, 2026c]: asks what from experience persists, and proposes selective retention/operator update, while stressing that experience itself is not a literal LoRA update.
4. **Readout Genesis Standalone Synthesis / MEMK** (July 2026) [Lahtee, 2026d]: builds a typed domain architecture of meaning–experience–memory–knowledge, and adds semantic potential / informational work / barrier crossing together with a claim firewall.
5. **From Problem to Hypothesis** (September 2026) [Lahtee, 2026e]: asks why some hypotheses/routes are more accessible than others, and separates reachability from accessibility by adding attraction and recent-path momentum.
6. **Epistemic Fusion or Tunnel** v7.1 → v8.1 (September 2026) [Lahtee, 2026f]: asks how Human and AI change each other's route within a session and what survives the session boundary; adds candidate status, D-R-A, amplification, retention (Repair 7: measuring return), and direction.
7. **Meaning Before Naming → Experience Is Meaning-Giving** [Lahtee, 2026g,h]: asks what happens before explicit naming/prompting; ultimately formalizes experience as meaning-giving, resonance as external–internal congruence, rhythm as structured recurrence, and transition readiness.
8. **Agency Potential → Potential as a Readout** (September 2026) [Lahtee, 2026i]: asks how the potential of agency can be measured without collapsing potential into observed performance; gives the R–D–X–F envelope over a two-layer witnessed set, with a lemma on bounded suppression and an identifiability certificate.
9. **Choice Begins Before Choice** [Lahtee, 2026j]: finds the missing interface between feasible capacity and overt choice, and sets up the Live Possibility Field as a dynamic pre-choice object.
10. **CTSA Human Return** [Lahtee, 2026k]: asks, at the far end, what genuinely comes back with the human once AI has been removed, and what gain/loss/regulation/provenance/warrant/direction that return has.

11. **After Labour** (6 September 2026) [Lahtee, 2026m]: asks what happens to human systemic position, not merely to employment, once productive intelligence and embodied production become reproducible through capital; adds labour centrality, the Citizen Claim Threshold, ownership accumulation, scarce-asset rent burden, aggregate-demand realization, social reproduction, and a typed Human Systemic Position audit index.
12. **The Human Conversion Imperative** (6 September 2026) [Lahtee, 2026n]: asks whether machine-capability growth is being converted into durable human capability rather than merely substituted for it; adds the Human Conversion Vector, conversion elasticities, a Bad-Mode state space, the Reversibility Window, and an Urgency Vector.
13. **How Humans Should Converse with AI** (Dialogue Conversion Protocol, v1.3, 6 September 2026) [Lahtee, 2026o]: asks which topic deserves to become the subject of a human-AI dialogue and who owns that choice before the eight-stage protocol begins; adds the Topic Entry Condition, the Problem-First Dialogue Principle, and closes the protocol through Action and World Feedback.
14. **From Assistance to Human Capability** (RG-HCA, v1.1 glosa-checked edition of the 6 September 2026 v1.0 original) [Lahtee, 2026p]: asks under what architecture AI may proactively advance human capability while preserving human-owned ends, distinguishing structural barriers from personal deficits, and separating assisted performance from durable Human Return; adds the Barrier Readout stage, endorsed-route non-collapse laws, a monotone assistance-fading rule, a world-side opportunity readout, and a conditional advancement estimand.

A relayed external reader's assessment of the Dialogue Conversion Protocol against world literature (2026-09-06, unverified: relayed from another AI session in chat, none of its own citations independently checked here) is recorded only as a readout, not adopted as validation. It placed the protocol's conceptual architecture and distinctiveness near the top of its own unvalidated scale while rating empirical status low pending experimental data, and it named several external comparanda (mixed-initiative interaction, problem framing, goal-centred collaboration, human–AI synergy meta-analysis, cognitive forcing, and AI-and-learning field experiments) as same/different/cited candidates for a future citation-lineage check, not as confirmed priors.

7 Epistemic Status of Each Segment

Table 2: Epistemic status of each segment of the river.

Level	What belongs at this level
Root / inherited architecture	Retained difference before semantic naming; ordered lineage; translation/readout/loss discipline; claims cannot borrow certainty.
Domain definitions	Meaning, experience, memory, semantic potential, informational work, live possibility set/profile, witnessed option set under a declared domain.
DR conceptual theses	Experience as meaning-giving; prompt as bounded articulation; choice downstream of live possibility; agency as a corrigible readout–action loop; two-layer potential (open conjecture).
OPEN empirical hypotheses	Incremental value of resonance; rhythm beyond exposure count; recent-path momentum; history-shaped accessibility; barrier/transition topology; Human–AI history effects beyond the current prompt.
Finite diagnostics / proved on a stated model	χ_{recip}, frozen session descendant counts, declared accessibility profiles, study-specific thresholds; bounded-suppression lemmas L1–L4 and the battery of *Potential as a Readout* (exact algebra on a declared model, not a theory about society).
Measurement architecture	Potential as a Readout, recoverable gaps, Human Return, CTSA, Y^{return}/RET, gain/loss/provenance/warrant/direction ledgers.
Normative/constitutive conditions	Difference, Resistance, retained human Agency under the definition of durable epistemic expansion; structural deprivation requires causal/counterfactual/normative support; NC-79 separates diagnosis from attribution.
New in v1.3 (After Labour, the Human Conversion Imperative, and the Dialogue Conversion Protocol)	
Domain definitions	Topic Entry (LiveProblem/OpenExploration/RoutineDelegation, (47)); the self-contained readout $\text{Readout}_{Q,O,c}(S)$ ((52)); four-dimensional labour centrality L_t ((53)); the Human Conversion Vector C_t^H ((61)); the Reversibility Window W_j ((63)); the Urgency Vector $U_{j,t}$ ((64)).
Accounting identities	The Citizen Claim Threshold q_t^{min} ((54)); the Human Systemic Position audit index P_t^H ((58)) and its non-collapse $\dot{Y}_t > 0 \not\Rightarrow \dot{P}_t^H > 0$ ((59)); the Human Return $\to$ Action $\to$ World Feedback closure ((50)–(51)) — bookkeeping/definitional results inside a declared stylized model, not validated causal or welfare laws.
OPEN empirical hypotheses	The Problem-First Dialogue Principle ((48)) and its non-collapse ((49)); the World-closure proposition; HCI's Reversibility Principle (underlying (63)–(64)); HCI's Conversion non-identity, Assistance-conversion divergence, and Distributional conversion propositions — all tagged OPEN in their own source papers, none upgraded here.
New in v1.4 (RG-HCA)	
Domain definitions	The Life-Capital Context Vector K^{life} ((68)); the typed barrier ledger B^{bar} ((69)); the endorsed-route set C^{live} ((71)); the world-side opportunity readout Ω^{real} ((75)).
Accounting identities / non-collapse laws	Observed Difficulty $\neq$ Skill Deficit ((70)); Proactive Suggestion $\neq$ Human Goal Ownership, Capability Advancement $\neq$ AI Goal Authority, Scaffolding $\neq$ Control ((72)); Credential $\neq$ Capability, Market Legibility $\neq$ Human Worth ((76)) — definitional guards inside a declared candidate domain, not validated psychological laws.
OPEN empirical hypotheses	RG-HCA's assistance-fading rule tying h^{decisive} to Stable Unaided Return ((73)); the Human Capability Advancement Principle — both explicitly OPEN in RG-HCA, neither upgraded here.
Measurement architecture	The net advancement record A_i^{HCA} ((77)) and the conditional estimand $\text{ATE}_{\text{HCA}}(x)$ ((78)), both RG-HCA's own [DEFINITION, MEASUREMENT] objects.

8 Conclusion: What Master Makes Visible

After reviewing the whole corpus, what the Master Equation River reveals is not that every paper says the same thing, but that each paper fills a missing layer of the same process. The overall picture is

$$\begin{aligned}&\text{Experience changes the reader;}\\&\text{the changed reader changes what can become possible;}\\&\text{what becomes possible changes choice;}\\&\text{choice changes action and the world;}\\&\text{the changed world is read again.}\end{aligned} \tag{45}$$

So the Human–AI Readout Programme, at this point, can be read as a theory programme of recursive human world-readout, meaning, retention, possibility, agency, and return. The point that must be strictly maintained is non-collapse: accessibility is not warrant, resonance is not truth, rhythm is not meaning, retention is not improvement, potential is not performance, observation is not capacity, and AI-assisted success is not Human Return.

Before freezing Master as a canonical paper, three further things should be done: (1) use the single canonical notation of this document (Section 4); (2) build a claim-status ledger for every arrow in equation (44); and (3) design an empirical programme that tests the incremental value of history, resonance, rhythm, the live field, and Human Return against simpler rival models.

9 Glosa Check Record (K0)

Version v1.0 was checked under the glosa methodology [Lahtee, 2026l] before deposit: an independent reviewer (an AI session different from the drafting session) read against all 11 original records and found 3 issues that blocked deposit — Fusion was cited as v7 while the deposited record is v8.1, and the retention-gate formula is not in v8.1; Agency Potential was cited as a primary source even though it has been superseded and was not deposited; RG-MEMK was cited as the source of (4) even though the equation is in the deposited record's own eq. 14 — plus items requiring correction (the R_H rename was not disclosed, there was no glossary, and DOIs were missing). All of these have been corrected in this edition. For every source cited by the table in Section 2, there is a citation card recording the URL actually fetched, the page, and one exact passage; each card was re-read by a headless session of the same model (independence class I1 on the glosa ladder, not I3), and the document's central claim (that the dependency closure is a spiral) has one claim card that was read by three adversarial roles (Falsifier, Hostile Reviewer, Source Auditor). The cards and reports live in the glosa repository (`records/lit/master-equation-river`, `projects/GLS-2026-003`). Knowledge status remains K0: timestamped, citable, not yet checked at independence class I2 or above.

AI-assistance disclosure. The structuring, source retrieval, passage extraction, and drafting of v1.1 were done together with an AI assistant under the author's direction; the problem, the questions, the choice of proposals, and the content of v1.0 that anchors this version belong to the author. No AI or vendor is an author. The author's own command lines are in the Blackbox Log (concept DOI 10.5281/zenodo.22302518, entries BBL-2026-09-06-140 to -142).

10 The One Equation Along the Whole Line

Since v1.4 was deposited, the programme's whole equation corpus (946 numbered equations across 40 deposited chapters, this document's own 79 among them, together with 123 further occurrences from the Readout Genesis domain registries, the Philosophy & Logic textbook and the solver arc: 1,069 mapped occurrences and 793 canonical entries in Toledo v1.0.0, counted from its registry file at the time of writing) has been coded into **Toledo** [Lahtee, 2026q], a permanent public registry in which every equation is either the Genesis root object itself or a coded reading of that root through a domain q_D. This section states the single root equation Toledo and *Collapse of the Master River* (`registry/COLLAPSE.md`, the programme's own mapping pass over Toledo's registry) read this whole river as an instance of, then reports which of this document's own segments already carry a Toledo code and which do not — adding no new claim and raising no tier above what `registry/CANONICAL.json` itself records.

10.1 The Weld

$$\delta_R = (a \natural b) \vdash_{Th_{coqc}} L_R = D_W - W \vdash_{Dr} F \text{ (MQ.08 stepper)} \equiv \{ q_D : q_D \circ F = F_D^{\#} \circ q_D \}.$$

This equation carries no number of its own — it is not a new claim added to (1)–(79), only the same root spine (CAN-002, CAN-201, CAN-001, CAN-003, CAN-006, CAN-007, CAN-008, CAN-004, CAN-009, below) restated once in the form `COLLAPSE.md` § 0 already gives it. Stating the master equation and admitting a domain are the same act: a retained distinction δ_R forces, under Th_coqc (machine-checked in a finite model), the graph Laplacian L_R and, under Dr (forced-but-not-independently-verified), the finite stepper F; a domain is nothing more than a translation q_D that this same stepper commutes with.

Nine coded objects in `CANONICAL.json` (ids CAN-002, CAN-201, CAN-001, CAN-003, CAN-006, CAN-007, CAN-008, CAN-004, CAN-009) together state this weld and the order in which it is read (world → retained state → readout → weld/retention → stepper → domain admission → quotient → non-collapse guard → semantic ordering → historical closure, then the next world-state). **Their own codes changed between the pass that produced `COLLAPSE.md` and the registry as it stands today**: `COLLAPSE.md` recorded all nine as `domain: "root"`; Toledo's current N3/N4 coding instead resolves each to a method-domain (M) reading of one of five literal Genesis root objects (`weld`, `EQ-015`, `EQ-002`, `A.5`, `A.8`) — the bare root itself is never a coded entry, only its readings are. Table 3 gives the current code for each, read directly from `CANONICAL.json`; no tier is raised in the re-coding, and CAN-001's own tier (Th_coqc for $\delta_R \vdash L_R$ only, Dr for the F-stepper) is unchanged.

Table 3: The nine-element root spine (`COLLAPSE.md` § 1), read against its current Toledo code.

CAN id	Key	Toledo code	Root	Tier	Status
CAN-002	`root-state-tuple`	`EQ-015/M.01.v1`	EQ-015	Definition	current
CAN-201	`root-readout-gate`	`EQ-002/M.03.v1`	EQ-002	Definition	current
CAN-001	`root-weld`	`weld/M.01.v1`	weld	Th_coqc / Dr	current
CAN-003	`root-stepper`	`EQ-015/M.02.v1`	EQ-015	untagged	current
CAN-006	`domain-weld`	`weld/M.02.v1`	weld	Definition	current
CAN-007	`reader-equivalence`	`weld/M.03.v1`	weld	Definition	current
CAN-008	`constitutional-noncollapse`	`A.5/M.01.v1`	A.5	Definition	current
CAN-004	`constitutional-ordering`	`EQ-015/M.03.v1`	EQ-015	Dr	current
CAN-009	`historical-invariance`	`A.8/M.01.v1`	A.8	Dr	current

10.2 Reading Table: Root × Domain

Table 4 counts, for each of the five literal roots underlying the spine above, how many canonical Toledo entries currently read that root through each of Toledo's eight domain letters (E epistemic, H human–AI, S social, W world-system, M method, P physics, C chemistry, B biology/health — `toledo/README.md`). A count of zero (shown —) is reported as exactly that: no code yet, not a gap this document fills. `weld`, the Social-Instability/Peace spine's own root (CAN-115, social reading of `weld`), has no chemistry, physics, or biology reading; `EQ-002` (the readout gate) has no social or biology reading; only `EQ-015` is read into physics (117) or biology (9), because that root is Genesis's own general stepper object, reused far outside this river's own social-justice-peace scope — reported here because it is true of the root this spine cites, not because Master River itself contains a physics or biology segment.

Table 4: Root × domain: count of Toledo canonical entries reading each literal root through each domain letter.

Root	E	H	S	W	M	P	C	B
`weld`	10	12	50	2	14	—	—	—
`EQ-015`	15	37	65	59	15	117	—	9
`EQ-002`	12	4	—	3	3	—	—	—

Table 4 continued

Root	E	H	S	W	M	P	C	B
`A.5`	9	19	20	20	32	—	—	—
`A.8`	3	4	1	—	19	—	—	—
Total	**49**	**76**	**136**	**84**	**83**	**117**	**0**	**9**

10.3 Which Root Each Segment of This River Reads

Read against `CANONICAL.json`'s own per-equation index (its `_n3_in_master_river` field, which names, for 55 of this document's 79 equations, the canonical code that equation is an occurrence of), the segments of Sections 3 to 5 and 8 read the five roots above as follows. Where no equation in a segment's range appears in that index, this is stated plainly rather than filled by inference.

- Section 3.1 (readout, meaning, experience; (1)–(5)) reads `EQ-002` ((1)) and `EQ-015` ((2)–(5)).
- Section 3.2 (naming as articulation and restructuring; (6)–(7)) reads `EQ-015`.
- Section 3.3 (retention and the changing future reader; (8)) reads `EQ-015`.
- Section 3.4 (resonance; (9)–(11)) reads `EQ-015`.
- Section 3.5 (rhythm, momentum, accessibility; (12)–(15)) reads `EQ-015` for (12)–(14); (15) has no Toledo code yet.
- Section 3.6 (accumulation, barrier, release; (16)–(18)) reads `EQ-015`.
- Section 3.7 (accessibility to the live possibility field; (19)–(24)) reads `A.5` throughout, with (21) also carrying an `EQ-015` reading.
- Section 3.8 (potential as a readout; (25)) reads `EQ-015`.
- Section 3.9 (power before choice, recoverable live-field gap; (26)) reads `EQ-015`.
- Section 3.10 (human–AI coupling before the prompt; (27)–(30)) reads `EQ-015` throughout, with (27) also carrying an `A.5` reading.
- Section 3.11 (within-session amplification, retention, direction; (31)–(40)) reads `EQ-015` at (31) and (38) only; (32)–(37) and (39)–(40) have no Toledo code yet.
- Section 3.12 (Human Return and CTSA; (41)–(42)) has no Toledo code yet.
- Section 4 (canonical notation corrections; (43)) has no Toledo code yet.
- Section 3.13 (topic entry and agenda ownership; (46)–(49)) has no Toledo code yet.
- Section 3.14 (Human Return, action, and world feedback; (50)–(51)) has no Toledo code yet.
- Section 3.15 (the world-system layer; (53)–(64)) reads `EQ-015` throughout; (52) has no Toledo code yet.
- Section 3.16 (barrier readout, endorsement, scaffold fading, opportunity conversion; (67)–(78)) reads `EQ-002` ((67)), `EQ-015` ((68), (73)–(74)), `A.5` ((69)–(72), (75)–(76)), and `A.8` ((77)–(78)).
- Sections 5 and 8 (the river-summary arrows (44), (45), (65), (66), and (79)) carry no code in `CANONICAL.json`'s per-equation index. `COLLAPSE.md` § 2 separately identifies (44) as a human–AI reading of CAN-004 and CAN-003 (now `EQ-015/M.03.v1` and `EQ-015/M.02.v1`), (66) as a reading of CAN-201 and CAN-006 (`EQ-002/M.03.v1`, `weld/M.02.v1`), and (79) as a reading of CAN-006 again; this is a hand-checked, second, narrower identification (`COLLAPSE.md`'s own "manual (verified textual match)" evidence tier) that the automated per-equation index does not yet carry, and both readings are reported here rather than one silently preferred over the other.

Of the nineteen segments listed above, **thirteen** carry at least one Toledo code (three of those thirteen — Sections 3.5, 3.11 and 3.15 — only for part of their own equation range) and **six** carry none in `CANONICAL.json`'s per-equation index: Section 3.12 (Human Return and CTSA), Section 4 (canonical notation corrections), Section 3.13 (topic entry), Section 3.14 (Human Return, action, and world feedback), Section 5 (the corrected master equation river), and Section 8 (the conclusion) — though, as noted above, `COLLAPSE.md` separately hand-checks three of Section 5's five river-summary equations against the root spine by its own narrower method. Appendix C gives the same reading equation-by-equation, for all 79.

v1.6 addition: every $\mathbb{Q}$-valued computation in this river reads the IDM number ladder. None of the equations above introduces a continuum object of its own — no completed limit, no $h \to 0$, no point of zero size is computed anywhere in this river; every scalar that is actually evaluated (χ_{recip}, η_s, g_n, the accessibility weights $\kappa^{\mathrm{sem}}/\lambda^{\mathrm{live}}$, the retention gate, the conversion percentiles) is a finite rational readout. This is not a new claim about this river; it is the same readout-first floor Information Discrete Mathematics states for any computed quantity, and it is itself coded into Toledo as the discrete number ladder $D \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{R}$ (Toledo root codes `D`, `Z`, `Q`, `R` — the continuum $\mathbb{R}$ is the readout of the discrete root, never silently injected) together with the weld/root reading $\delta_R \vdash_{Th_{coqc}} L_R$ already cited in Section 10.1 (Toledo code `weld/M.01.v1`, CAN-001). Full treatise: DOI 10.5281/zenodo.22644131 [Lahtee, 2026r]. No tier in this river is raised by this observation; it only names, for the reader, which finite-mathematics floor every $\mathbb{Q}$-valued object above already stands on.

Appendix A. Symbol Glossary

Table 5: Symbol glossary.

Symbol	Meaning	Defined in
x_n	Phenomenon / world-side encounter at step n.	[Lahtee, 2026d,h]
$r_n = R_H(\cdot)$	Finite human readout of x_n.	[Lahtee, 2026d]
H_n	Retained human state (the reader's history-bearing operator).	[Lahtee, 2026c,h]
c_n, Q_n	Context; the criterion/question that governs the reading.	[Lahtee, 2026d,e]
μ_n, Ψ_H	The meaning-giving relation and its operator.	[Lahtee, 2026h]
E_n, Φ_E	Experience and the experience operator (MEMK).	[Lahtee, 2026d], eq. 14
γ_n^μ	Meaning strength (renamed from MEMK's κ_n).	Section 4.1
ℓ_n, L_H	Lexical naming / articulation operator; L_H is also used as the transport from H_t to Q_t.	[Lahtee, 2026g,h]
U_H, Retain	State-update and selective retention.	[Lahtee, 2026c]
$M_{H,n}$, $I_{H,n}$, C_H	Retained memory organization; retrieved internal organization; congruence measure.	[Lahtee, 2026h]
T_n, B_n, Ω	Encounter history; boundary/structure parameters; rhythm operator.	[Lahtee, 2026h]
m_t, $a_{Q,t}$, $c_{A,t,Q}$, ξ_t	Recent-path momentum; attraction; switching cost; residual.	[Lahtee, 2026e]
κ^{sem}	Semantic-route accessibility.	[Lahtee, 2026e], Section 4.1
V_{sem}, W_{info}, $\Delta V^\dagger$	Semantic potential; informational work; activation barrier (not physical energy).	[Lahtee, 2026d,h]
$\Pi^{\text{phys}}, \Pi^{\text{feas}}, \Pi^{\text{live}}, \Pi^{\text{wit}}$	Physically possible / structurally feasible / experientially live / witnessed policy sets.	[Lahtee, 2026j,i]
λ^{live}, $L_{A,t}(g)$, τ_{live}	Live-policy accessibility; the Live Possibility Field; live threshold.	[Lahtee, 2026j]
g, A, z, h, T, B, P	Task; agent; conditions; history; horizon; budget; declared model.	[Lahtee, 2026i]
$\text{Read}_g, D_g, X_g, F_g$	Read / actor-directed decision / causal enactment / feedback-objection events.	[Lahtee, 2026i]
$p^*_{A,g}$, J_{feas}, D_L	The potential envelope; the feasible intervention set; live-field distance.	[Lahtee, 2026i,j]
$Y_{\text{obs}} = \mathscr{O}_q(H_{0:T})$	The evaluator's instrument readout of the trajectory.	[Lahtee, 2026i,j]
Q_t, Y_t, $E^{\text{AI}}_{H,t}$	Prompt; AI output; the human's experience of that output.	[Lahtee, 2026f,h]
Z_{dlg}, u_H, u_{AI}, χ_{recip}, D_{recip}, Σ	Dialogue state; human/AI inputs; reciprocal-lineage ratio; reciprocal descendants; all descendants.	[Lahtee, 2026f]
η_s, Ω_s^H, B_sA_s, d_H	Session retention gate; human operator state; rank-restricted candidate update; state dimension.	[Lahtee, 2026f] (gate); Section 3.11 (proposal)
Y^{return}, RET, AUG, SYN, J_s^*	Unaided return battery; return / augmentation / synergy estimands; three-axis outcome.	[Lahtee, 2026f] Repair 7; [Lahtee, 2026i]
$K_{\text{like}}, K_{\text{validated}}$	Candidate (knowledge-like) vs. validated status.	[Lahtee, 2026a,f]
Δ_s, G_s, T_s, P_H	Expansion–tunnel balance; gain; tunnel; human performance.	[Lahtee, 2026f]
H_{return} ($= R_H$ in CTSA)	Human Return audit record $\langle G_{\text{CTSA}}, L, M, P, W, \Delta_{\text{dir}} \rangle$.	[Lahtee, 2026k], Section 4.4
New in v1.3		
TopicEntry, $P^{\text{live}}_{H,t}$	Topic-selection variable (LiveProblem/OpenExploration/RoutineDelegation); a live problem, a presently consequential unresolved difference.	[Lahtee, 2026o], (47)
I_s, a_s, $\delta^{\text{world}}_{s+1}$	DCP's integration record; human-selected next action; the returned world record.	[Lahtee, 2026o], (50)
$\text{Readout}_{Q,O,c}(S) = z$	After Labour's self-contained readout definition ($z \neq S$); not the root readout R_H used elsewhere in this river.	[Lahtee, 2026m], (52)
L_t	Four-dimensional labour centrality: task, income, bottleneck, bargain.	[Lahtee, 2026m], (53)
$q_t^{\min}$, Γ_t, Γ_t^{eff}	Citizen Claim Threshold; broad claim on output; effective material claim after rent/cross-border burdens.	[Lahtee, 2026m], (54)
$A^{\text{corr}}_{H,i,t}(g)$, $R^{\text{return}}_{H,t}$	World-system-level corrigible-agency envelope; delayed unaided return profile (After Labour's own restatement, kept typographically distinct from H_{return}).	[Lahtee, 2026m], (56)–(57)
P_t^H	Human Systemic Position audit index.	[Lahtee, 2026m], (58)
C_t^H, $\eta^{HC}_{j,t}$, M_t	Human Conversion Vector; local conversion elasticity per dimension; declared machine-capability measure.	[Lahtee, 2026n], (61)–(62)
W_j, $U_{j,t}$	Reversibility Window; Urgency term for dimension j.	[Lahtee, 2026n], (63)–(64)
New in v1.4		

Table 5 continued

Symbol	Meaning	Defined in
$K_{i,n}^{\text{life}}$	Life-Capital Context Vector: economic, foundational-literacy, language, digital, discretionary-time, health, mobility, mentor/network, and credential coordinates for agent i.	[Lahtee, 2026p], (68)
$B_{i,n}^{\text{bar}}$	Typed barrier ledger (Knowledge, Skill, Language, Tool, ResourceTime, Network, Credential, Permission, Opportunity, Unknown).	[Lahtee, 2026p], (69)
$C_{i,n}^{\text{cand}}$, $C_{i,n}^{\text{live}}$, Endorse_i	Candidate advancement routes; the human-endorsed subset; the endorsement predicate.	[Lahtee, 2026p], (71)
h^{decisive}	Level of decisive AI assistance/scaffolding subject to fading.	[Lahtee, 2026p], (73)
$Z_{\text{HCA},i,n}$	RG-HCA's candidate HCA domain state for agent i at step n (not asserted identical to the retained root state S_n).	[Lahtee, 2026p], (74)
$\Omega_{i,n}^{\text{real}}$, $\text{Cred}_{i,n}$, $\text{Net}_{i,n}$, $\text{Perm}_{i,n}$	World-side opportunity readout and its credential/network/permission arguments.	[Lahtee, 2026p], (75)
A_i^{HCA}, $\text{ATE}_{\text{HCA}}(x)$	Net advancement record; conditional average treatment estimand given a baseline life-context stratum x.	[Lahtee, 2026p], (77)–(78)

Appendix B. Th_coqc Ledger: Coq Formalisation of Equations (1)–(79)

Every numbered equation of this document is formalised in the Coq 8.20 development *Master Equation River — Coq formalisation of every equation*, v1.0, Zenodo software record DOI 10.5281/zenodo.22518450 [Lahtee, 2026s], on finite, discrete models only (nat, $\mathbb{Z}$, $\mathbb{Q}$, bool, finite lists and records; no `Coq.Reals`, no classical axioms, no `Admitted`, no top-level `Axiom`/`Parameter`). Each equation sits at exactly one tier. **Th_coqc**: a lemma proved in a stated finite model, reporting *Closed under the global context* under `Print Assumptions` (45 identifiers, 0 failed, `verify.sh`); non-collapse laws are proved by exhibiting a finite model in which the two readouts differ, so the lemma witnesses the possibility of divergence, not a claim about every model. **Definition**: the object is typed as the paper writes it. **Open**: the hypothesis is stated as a `Prop` and not proved. Where a proved lemma is only a narrow witness of a thesis this document keeps at tier Dr (equations 5, 8, 16), the ledger says so; the formalisation checks the internal consistency of the stated discrete models, not empirical truth. Continuum-looking expressions in the text (`exp`, ∂ ln, a maximum over an infinite set) are replaced in Coq by declared discrete counterparts and the replacement is recorded in the source comments.

v1.5 pointer update. The standalone software record above, DOI 10.5281/zenodo.22518450 [Lahtee, 2026s], is **obsoleted by Toledo v1.0.0**, DOI 10.5281/zenodo.22548770 (concept DOI 10.5281/zenodo.22537318) [Lahtee, 2026q]: these equations' Coq development now lives inside Toledo's own repository, one file per canonical code (the file-name mangling is the code itself, e.g. `weld/M.01.v1` → `weld__M_01_v1.v`), alongside a machine-checked build+`Print Assumptions` report for the whole registry. This is a location update only: the ledger below (Table 6) is the ledger as v1.4 deposited it, unchanged, and no tier or assumption cell in it is altered by this pointer.

v1.6 pointer update. Toledo itself has since advanced to v1.5.0, DOI 10.5281/zenodo.22642109 (same concept DOI 10.5281/zenodo.22537318) [Lahtee, 2026q]; the citation above and throughout this document now reads that record. This is again a location/version update only, carrying forward the same one-file-per-canonical-code Coq development; no tier or assumption cell in Table 6 is altered by this pointer, and neither is Appendix C's own reading of `CANONICAL.json` as it stood at build time.

Table 6: Th_coqc ledger (abridged from `MR_Ledger.md` in the Coq record).

Eq.	Module	Identifier(s)	Tier	Assumptions
1	MR_Foundation.v	R_H (Variable, Section Foundation1)	Definition	n/a (Definition, not a proof o
2	MR_Foundation.v	Psi_H (Variable, Section Foundation1)	Definition	—
3	MR_Foundation.v	MeaningModes (Record); meaning_modes_decomposition_faithful	Definition	closed
4	MR_Foundation.v	Phi_E (Variable, Section Foundation1)	Definition	—
5	MR_Foundation.v	eq5_experience_is_phenomenon_and_meaning_jointly	Th_coqc (narrow witness lemma only: a finite model where the compression is jointly sensitive to phenomenon and meaning; the paper's thesis itself stays at its own tier, Dr)	closed
6	MR_Foundation.v	L_H (Variable, Section Naming); witness naming_can_remain_unstable	Definition	closed
7	MR_Foundation.v	NamingChainStep (Record); naming_chain (Fixpoint)	Definition	—
8	MR_Foundation.v	eq8_retention_can_change_the_reader	Th_coqc (narrow witness lemma only: a finite model where retention-with-update is not idle; the paper's thesis stays Dr)	closed
9	MR_Resonance.v	Retrieve (Variable, Section Resonance)	Definition	—
10	MR_Resonance.v	C_H (Variable, Section Resonance)	Definition	—
11	MR_Resonance.v	eq11_resonance_non_collapse (+ notion_value_injective)	Th_coqc	closed
12	MR_Resonance.v	EncounterHistory, Omega (Variable) …	Definition	—
13	MR_Resonance.v	momentum (Fixpoint, scaffold); Open_eq13	Open	n/a (Open Prop, not proved by
14	MR_Resonance.v	accessibility_score, disc_gain_nat …	Open	n/a (Open Prop, not proved by
15	MR_Resonance.v	eq15_rhythm_alone_does_not_determine_accessibility, eq15_kappa_next_genuinely_varies	Th_coqc	closed
16	MR_Resonance.v	accum_work, accum_work_cons …	Th_coqc (bookkeeping facts about the definition only: monotone/non-negative sums over Q; the transition claim of eq. 18 remains Open)	closed
17	MR_Resonance.v	effective_work (Section EffectiveWork)	Definition	—
18	MR_Resonance.v	threshold_crossed, threshold_crossed_decidable; Open_eq18	Open	closed
19	MR_Live.v	Pi_live, eq19_live_subset_feas …	Th_coqc	closed
20	MR_Live.v	live_field	Definition	—
21	MR_Live.v	live_ge_threshold, Pi_live	Definition	—
22	MR_Live.v	is_valid_choice	Definition	—

Table 6 continued

Eq.	Module	Identifier(s)	Tier	Assumptions
23	MR_Live.v	eq23_enactment_may_differ_from_choice; Trajectory, Obs . . .	Th_coqc	closed
24	MR_Live.v	Stage, stage_value . . .	Th_coqc	closed
25	MR_Live.v	p_star; p_star_upper_bound	Definition	closed
26	MR_Live.v	live_field_gap	Definition	—
27	MR_Prompt.v	L_H (Variable, Section PrePromptCoupling)	Definition	—
28	MR_Prompt.v	AI_step, R_H_readout . . .	Definition	—
29	MR_Prompt.v	U_H (Variable), next_state	Definition	—
30	MR_Prompt.v	toy_update, toy_lambda_live . . .	Th_coqc	closed
31	MR_Prompt.v	F_hash_dlg (Variable), dlg_step	Definition	—
32	MR_Prompt.v	chi_recip, eq32_chi_recip_bounds	Th_coqc	closed
33	MR_Prompt.v	ReturnBattery, mk_return_battery	Definition	—
34	MR_Prompt.v	RET, eq34_RET_rearrangement	Th_coqc	closed
37	MR_Retention.v	ai_momentum_resets	Definition	—
38	MR_Retention.v	KnowledgeStatus, status_value . . .	Th_coqc	closed
39	MR_Retention.v	Delta_gt, eq39_sign_scale_pos_case . . .	Th_coqc	closed
40	MR_Retention.v	AUG, SYN . . .	Th_coqc	closed
41	MR_Retention.v	HReturn, mk_h_return	Definition	—
42	MR_Retention.v	EndChainNotion, end_chain_value . . .	Th_coqc	closed
43	MR_Corrections.v	gamma_mu, kappa_sem . . .	Definition	—
44	MR_River.v	master_river_44, one_pass_to_h_return . . .	Definition	—
45	MR_River.v	closing_summary_45	Definition	—
65	MR_River.v	river_tail_65	Definition	—
66	MR_River.v	outer_loop_66	Definition	—
46	MR_TopicEntry.v	L_H (Variable, Section TopicEntryTransport); topic_entry_transport	Definition	—
47	MR_TopicEntry.v	TopicEntry (Inductive); Legitimate	Definition	—
48	MR_TopicEntry.v	Open_eq48	Open	n/a (Open Prop, not proved by
49	MR_TopicEntry.v	ProblemOnlyPolicy (Definition, scaffolding only); Open_eq49	Open	n/a (Open Prop, not proved by
50	MR_TopicEntry.v	dcp_closure_50 (Section HumanReturnClosure)	Definition	—
51	MR_TopicEntry.v	CycleStage (Inductive); cycle_next_51	Definition	—
52	MR_WorldSystem.v	readout_52 (Variable, Section ReadoutDefinition); Hypothesis readout_ne_state; readout_52_hypothesis_satisfiable_on_bool	Definition	closed
53	MR_WorldSystem.v	LabourCentrality; mk_labour_centrality	Definition	—
54	MR_WorldSystem.v	q_min; eq54_citizen_claim_threshold_identity	Th_coqc	closed
55	MR_WorldSystem.v	Pi_live_ws; eq55_live_subset_feas_ws, eq55_live_subset_phys_ws . . .	Th_coqc	closed
56	MR_WorldSystem.v	corrigible_agency_ws; corrigible_agency_ws_upper_bound	Definition	closed
57	MR_WorldSystem.v	ReturnProfileWS; mk_return_profile_ws	Definition	—
58	MR_WorldSystem.v	Qpow_nat (Fixpoint); P_H_index	Definition	—
59	MR_WorldSystem.v	ddiff; eq59_output_rise_not_position_rise	Th_coqc	closed
60	MR_WorldSystem.v	eq60_machine_expansion_not_human_expansion	Th_coqc	closed
61	MR_WorldSystem.v	HumanConversionVector; mk_human_conversion_vector	Definition	—
62	MR_WorldSystem.v	eta_HC (Section ConversionElasticity)	Definition	—
63	MR_WorldSystem.v	in_reversibility_window, reversibility_window	Definition	—
64	MR_WorldSystem.v	urgency_term	Definition	—
67	MR_HCA.v	hca_river_67 (Section HCARiver)	Definition	—
68	MR_HCA.v	LifeCapitalContext; mk_life_capital_context	Definition	—
69	MR_HCA.v	BarrierType (Inductive); all_barrier_types; BarrierLedger	Definition	—
70	MR_HCA.v	eq70_observed_difficulty_not_skill_deficit	Th_coqc	closed
71	MR_HCA.v	C_live (Section EndorsedRoutes); C_live_subset_C_cand	Definition	closed
72	MR_HCA.v	eq72a_proactive_suggestion_not_human_goal_ownership, eq72b_capability_advancement_not_ai_goal_authority . . .	Th_coqc	closed
73	MR_HCA.v	hca_ddiff (scaffold); Open_eq73	Open	n/a (Open Prop, not proved by
74	MR_HCA.v	world_closure_74 (Section WorldClosureHCA)	Definition	—
75	MR_HCA.v	omega_real_75 (Section OpportunityReadout)	Definition	—
76	MR_HCA.v	eq76a_credential_not_capability, eq76b_market_legibility_not_human_worth	Th_coqc	closed
77	MR_HCA.v	NetAdvancementRecord; mk_net_advancement_record	Definition	—
78	MR_HCA.v	stratum_members, qsum . . .	Definition	—
79	MR_HCA.v	hca_outer_layer_79 (Section HCAOuterLoopLayer)	Definition	—

APPENDIX C. CANONICAL REGISTRY OF EQUATIONS (1)–(79)

This table reads (1)–(79) of this document against `registry/CANONICAL.json`'s own `_n3_in_master_river` field — the field Toledo's N4 registry build uses to record which of its 793 canonical entries is an occurrence of an equation in *this* document, and which equation number(s) it occurs at. One row per equation number; where an equation is the occasion for more than one canonical code (the paper's own (27), for instance, is a single displayed transport that Toledo separately codes as an `EQ-015` reading, an `A.5` reading, and a second `EQ-015` reading), every code is listed. Tier and status are copied verbatim from `CANONICAL.json`; a SPLIT status means the code is retired in Toledo's own registry in favour of several finer-grained child codes (a later N4 correction, not a change made by this document) — the code is still shown here because it is what the field currently names. Read plainly: **55 of these 79 equations carry at least one Toledo code; 24 do not**. An equation lacking a code here is not thereby false or lower-tier than stated in the body of this document — it means only that Toledo's own build has not yet coded that occurrence, exactly the honesty Section 10 already discloses at the segment level.

v1.6 addition. For each of the twenty-four rows that carried no code, this edition now names a specific registrar proposal instead of leaving the row unmapped: `registry/proposals/master_river_v1_6.json` either proposes a new occurrence against an already-existing Toledo entry found equivalent under the phi-criterion (renaming / positive scale / constant substitution), or, for three constructs with no existing match, a wholly new reading entry. No code below is asserted by this document itself — `code pending registrar` means exactly that: the chair fills in the final Toledo code once the registrar merges (or declines) the named proposal, and this table is updated again only then.

Table 7: Canonical registry of equations (1)–(79): Toledo code, root, tier, and status, read from `CANONICAL.json`.

Eq.	Toledo code	Root	Tier	Status
1	`EQ-002/E.01.v1`	EQ-002	Definition	current
2	`EQ-015/E.01.v1; EQ-015/E.03.v1`	EQ-015; EQ-015	Definition; Definition	current; current
3	`EQ-015/E.01.v1`	EQ-015	Definition	current
4	`EQ-015/E.04.v1`	EQ-015	Definition	current
5	`EQ-015/E.04.v1`	EQ-015	Definition	current
6	`EQ-015/E.05.v1`	EQ-015	Definition	current
7	`EQ-015/E.05.v1`	EQ-015	Definition	current
8	`EQ-015/E.08.v1`	EQ-015	Definition	current
9	`EQ-015/E.06.v1`	EQ-015	Dr	current
10	`EQ-015/E.06.v1`	EQ-015	Dr	current
11	`EQ-015/E.06.v1`	EQ-015	Dr	current
12	`EQ-015/M.08.v1`	EQ-015	Definition	current
13	`EQ-015/M.08.v1`	EQ-015	Definition	current
14	`EQ-015/M.08.v1`	EQ-015	Definition	current
15	`EQ-015/M.08.v1`	EQ-015	Definition	current
16	`EQ-015/E.07.v1`	EQ-015	Definition	current
17	`EQ-015/E.07.v1`	EQ-015	Definition	current
18	`EQ-015/E.07.v1`	EQ-015	Definition	current
19	`A.5/S.03.v1`	A.5	Definition	split
20	`A.5/S.03.v1`	A.5	Definition	split
21	`EQ-015/H.07.v1; A.5/S.03.v1`	EQ-015; A.5	Definition; Definition	current; split
22	`A.5/S.03.v1`	A.5	Definition	split
23	`A.5/S.03.v1`	A.5	Definition	split
24	`A.5/S.03.v1`	A.5	Definition	split
25	`EQ-015/S.06.v1`	EQ-015	Definition	split
26	`EQ-015/S.08.v1`	EQ-015	Definition	current
27	`EQ-015/H.01.v1; A.5/H.01.v1; EQ-015/H.03.v1`	EQ-015; A.5; EQ-015	Definition; Definition; Definition	current; current; current
28	`EQ-015/H.01.v1; EQ-015/H.03.v1`	EQ-015; EQ-015	Definition; Definition	current; current
29	`EQ-015/E.08.v1; EQ-015/H.03.v1`	EQ-015; EQ-015	Definition; Definition	current; current
30	`EQ-015/H.03.v1`	EQ-015	Definition	current
31	`EQ-015/H.05.v1`	EQ-015	Definition	current
32	`EQ-015/H.05.v1`	EQ-015	Definition	current
33	`weld/H.07.v1`	weld	Definition	current
34	`weld/H.07.v1`	weld	Definition	current
35	`weld/H.07.v1`	weld	Definition	current
36	`weld/H.07.v1`	weld	Definition	current
37	`weld/H.38.v1`	weld	Definition	current
38	`EQ-015/H.10.v1`	EQ-015	Definition	current
39	`EQ-015/H.12.v1`	EQ-015	Dr	current
40	`EQ-015/H.19.v1`	EQ-015	Definition	current
41	`EQ-015/H.16.v1`	EQ-015	Definition	current
42	`A.5/H.09.v1`	A.5	Definition	current
43	`EQ-015/E.16.v1`	EQ-015	Definition	current
44	`EQ-015/M.03.v1`	EQ-015	Dr	current
45	`EQ-015/M.16.v1`	EQ-015	Definition	current
46	`EQ-015/H.03.v1`	EQ-015	Definition	current
47	`A.5/H.02.v1`	A.5	Definition	current
48	`A.5/H.02.v1`	A.5	Definition	current
49	`A.5/H.02.v1`	A.5	Definition	current

Table 7 continued

Eq.	Toledo code	Root	Tier	Status
50	`EQ-015/H.24.v1`	EQ-015	Definition	current
51	`EQ-015/H.24.v1`	EQ-015	Definition	current
52	`EQ-002/M.03.v1`	EQ-002	Definition	current
53	`EQ-015/W.04.v1`	EQ-015	Definition	split
54	`EQ-015/W.05.v1`	EQ-015	Definition	split
55	`EQ-015/H.07.v1`	EQ-015	Definition	current
56	`EQ-015/W.14.v1`	EQ-015	Definition	current
57	`EQ-015/W.15.v1`	EQ-015	Definition	current
58	`EQ-015/W.10.v1`	EQ-015	Definition	split
59	`EQ-015/W.10.v1`	EQ-015	Definition	split
60	`EQ-015/W.16.v1`	EQ-015	Definition	split
61	`EQ-015/W.16.v1`	EQ-015	Definition	split
62	`EQ-015/W.16.v1`	EQ-015	Definition	split
63	`EQ-015/W.19.v1`	EQ-015	Definition	split
64	`EQ-015/W.19.v1`	EQ-015	Definition	split
65	`EQ-015/H.24.v1`	EQ-015	Definition	current
66	`EQ-002/M.03.v1`	EQ-002	Definition	current
67	`EQ-002/H.03.v1`	EQ-002	Definition	current
68	`EQ-015/H.31.v1`	EQ-015	Definition	current
69	`A.5/H.16.v1`	A.5	Definition	current
70	`A.5/H.16.v1`	A.5	Definition	current
71	`A.5/H.17.v1`	A.5	Definition	current
72	`A.5/H.17.v1`	A.5	Definition	current
73	`EQ-015/H.32.v1`	EQ-015	Open	current
74	`EQ-015/H.24.v1`	EQ-015	Definition	current
75	`A.5/H.18.v1`	A.5	Definition	current
76	`A.5/H.18.v1`	A.5	Definition	current
77	`A.8/H.03.v1`	A.8	Definition	current
78	`A.8/H.03.v1`	A.8	Definition	current
79	`weld/M.02.v1`	weld	Definition	current

A separate, hand-checked identification exists for three of the twenty-four unmapped equations: `COLLAPSE.md` § 2 reads (44) as a human–AI reading of `EQ-015/M.03.v1` and `EQ-015/M.02.v1` (CAN-004, CAN-003), (66) as a reading of `EQ-002/M.03.v1` and `weld/M.02.v1` (CAN-201, CAN-006), and (79) as a further reading of `weld/M.02.v1` (CAN-006) — by direct textual comparison rather than the automated field this table reads, and disclosed at `COLLAPSE.md`'s own "manual (verified textual match)" evidence tier, not a higher one. (65) and the remaining twenty unmapped equations ((15), (32)–(37), (39)–(43), (45)–(52)) carry no code and no such hand-check under either source consulted for this appendix.

v1.6 addition: registrar proposals for all twenty-four. `registry/proposals/master_river_v1_6.json` carries the three `COLLAPSE.md` hand-checks above forward unchanged as occurrence proposals (`MR:eq.44`, `MR:eq.66`, `MR:eq.79`), and adds a phi-criterion occurrence proposal for eighteen more of the remaining twenty-one (`MR:eq.15`, `eq.32`–`eq.36`, `eq.39`–`eq.42`, `eq.46`–`eq.52`, `eq.65`), plus one new-reading proposal each for the three constructs no existing Toledo entry was found to match under that criterion (`eq.37`, the AI-side session-momentum reset; `eq.43`, the three-way κ notation rename; `eq.45`, the river's closing narrative restatement). This is a proposal only: no code above is asserted by this document, and the registrar may merge, revise, or decline any of the twenty-four.

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